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VARIETY EVASIVE SUBSPACE FAMILIES

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Abstract. We introduce the problem of constructing explicit *variety evasive subspace families*. Given a family $\mathcal{F}$ of subvarieties of a projective or affine space, a collection $\mathcal{H}$ of projective or affine k-subspaces is $(\mathcal{F}, \varepsilon)$-*evasive* if for every $\mathcal{V} \in \mathcal{F}$, all but at most ε-fraction of $W \in \mathcal{H}$ intersect every irreducible component of $\mathcal{V}$ with (at most) the expected dimension. The problem of constructing such an explicit subspace family generalizes both deterministic black-box polynomial identity testing (PIT) and the problem of constructing explicit (weak) lossless rank condensers.

Using Chow forms, we construct explicit k-subspace families of polynomial size that are evasive for all varieties of bounded degree in a projective or affine n-space. As one application, we obtain a complete derandomization of Noether's normalization lemma for varieties of low degree in a projective or affine n-space. In another application, we obtain a simple polynomial-time black-box PIT algorithm for depth-4 arithmetic circuits with bounded top fan-in and bottom fan-in that are not in the Sylvester–Gallai configuration, improving and simplifying a result of Gupta (ECCC TR 14-130).

As a complement of our explicit construction, we prove a tight lower bound for the size of k-subspace families that are evasive for degree-d varieties in a projective n-space. When $n - k = n^{\Omega(1)}$, the lower bound is superpolynomial unless d is bounded. The proof uses a dimension counting argument on Chow varieties that parametrize projective subvarieties.

Keywords. algebraic complexity, dimension reduction, Noether normalization, polynomial identity testing, pseudorandomness, varieties

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1. Introduction

Polynomial identity testing (PIT) is a fundamental problem in the areas of derandomization and algebraic complexity theory. The problem asks whether a multivariate polynomial, computed by an arithmetic circuit, formula, or other algebraic computational models, is identically zero. For example, the polynomial $(X+Y)(X-Y)-X^2-Y^2$ is identically zero while $(X+Y)^2-X^2$ is not.

It is easy to solve PIT in randomized polynomial time, as we may simply evaluate the input polynomial at a random point and check if the evaluation is zero. On the other hand, finding a deterministic polynomial-time PIT algorithm for general arithmetic circuits is a long-standing open problem. Such algorithms are known for some special cases, and we refer the readers to the surveys Saxena (2009, 2013); Shpilka & Yehudayoff (2010) for details.

Black box PIT algorithms are a special kind of PIT algorithm. A (deterministic) black-box PIT algorithm tests if a polynomial in a family $\mathcal{F}$ is zero by constructing a *hitting set* for $\mathcal{F}$, which is a finite collection $\mathcal{H}$ of evaluation points with the following property: for any nonzero $Q \in \mathcal{F}$, there exists $p \in \mathcal{H}$ such that the evaluation of Q at p is nonzero. After constructing a hitting set $\mathcal{H}$ for $\mathcal{F}$, the algorithm simply checks if the evaluation of the given polynomial at every point in $\mathcal{H}$ is zero. The problem of designing a deterministic black-box PIT algorithm for polynomials in $\mathcal{F}$ is thus equivalent to deterministically constructing a hitting set for $\mathcal{F}$. To make the algorithm efficient, such a hitting set should be small and efficiently computable. Hitting sets are also called *correct test sequences* and are studied in Heintz & Schnorr (1980); Pardo & Sebastián (2022).

From a geometric perspective, an n-variate nonzero polynomial Q over an algebraically closed field $\mathbb{F}$ defines a *hypersurface* $\mathcal{V}(Q) := \{\alpha \in \mathbb{F}^n : Q(\alpha) = 0\}$ of $\mathbb{F}^n$. A hitting set $\mathcal{H}$ for $\mathcal{F}$ has the property that for every nonzero $Q \in \mathcal{F}$, there exists a point $p \in \mathcal{H}$ that is disjoint from the hypersurface $\mathcal{V}(Q)$, or we say p *evades* $\mathcal{V}(Q)$. It is natural to consider the generalization of this property to higher dimensions/codimensions. Namely, we want to construct a finite collection $\mathcal{H}$ of *affine k-subspaces* (i.e., affine subspaces of dimension k) such that for every *variety* $\mathcal{V} \subseteq \mathbb{F}^n$ (i.e., solution set of a set of polynomial equations) from a certain family, some (or

most) $W \in \mathcal{H}$ *evade* $\mathcal{V}$, in the sense that the dimension of the intersection $\mathcal{V} \cap W$ is bounded by the expected dimension achieved by W in general position. A similar property can be defined for projective k-spaces, to be defined below. We call such a collection $\mathcal{H}$ of projective or affine k-subspaces a *variety evasive subspace family*. The formal definition is given below.

1.1. Variety Evasive Subspace Families. Let $\mathbb{F}$ be an algebraically closed field. An *affine n-space* $\mathbb{A}^n$, as a set, is simply defined to be the vector space $\mathbb{F}^n$. We also need the notion of a *projective n-space*, denoted by $\mathbb{P}^n$, which is (intuitively) the set of lines passing through the origin $\mathbf{0}$ of $\mathbb{A}^{n+1}$. Formally, it is defined to be the quotient set $(\mathbb{A}^{n+1} \setminus \{\mathbf{0}\})/\sim$, where $\sim$ is the equivalence relation defined by scaling, i.e., $u \sim v$ if $u = cv$ for some nonzero scalar $c \in \mathbb{F}$.

An *(affine) subvariety* $\mathcal{V} \subseteq \mathbb{A}^n$ is the set of common zeros of a set of n-variate polynomials over $\mathbb{F}$. Similarly, a *(projective) subvariety* $\mathcal{V} \subseteq \mathbb{P}^n$ is the set of common zeros of a set of *homogeneous* $(n+1)$-variate polynomials over $\mathbb{F}$, where we represent each element of $\mathbb{P}^n$ as an $(n+1)$-tuple in $\mathbb{A}^{n+1}$. In this paper, a *variety* refers to a subvariety of a projective or affine space, and is said to be *irreducible* if it cannot be written as a union of finitely many proper subvarieties. (Varieties in this paper are not necessarily irreducible. They are often called *algebraic sets* in literature.)

The *dimension* of a variety $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is intuitively the "degree of freedom" of picking a point in the variety. See Section 2.3 for its formal definition. In particular, for a linear subspace $V \subseteq \mathbb{A}^n$, the dimension of V as a variety is simply its linear-algebraic dimension.

For two irreducible subvarieties $\mathcal{V}_1$ and $\mathcal{V}_2$ of $\mathbb{P}^n$ or $\mathbb{A}^n$ in *general position*, we expect the dimension of $\mathcal{V}_1 \cap \mathcal{V}_2$ to be $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n$ (unless $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) < n$, in which case we expect $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$). The following definition captures the condition that $\dim(\mathcal{V}_1 \cap \mathcal{V}_2)$ is bounded by the expected dimension.

Definition 1.1 (Evading). *Let $\mathcal{V}_1$ and $\mathcal{V}_2$ be irreducible subvarieties of $\mathbb{P}^n$ or $\mathbb{A}^n$. We say* $\mathcal{V}_1$ evades $\mathcal{V}_2$ *if*

$$\dim(\mathcal{V}_1 \cap \mathcal{V}_2) \leq \dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n,$$

where the dimension of an empty set is assumed to be $-\infty$*. In particular, if* $\dim(\mathcal{V}_1)+\dim(\mathcal{V}_2) < n$*, then* $\mathcal{V}_1$ *evades* $\mathcal{V}_2$ *iff* $\mathcal{V}_1 \cap \mathcal{V}_2 = \emptyset$*.*

More generally, suppose $\mathcal{V}_1$ *is irreducible but* $\mathcal{V}_2$ *is possibly reducible. We say* $\mathcal{V}_1$ evades $\mathcal{V}_2$ *if it evades every irreducible component of* $\mathcal{V}_2$*.*

Next, we define *subspace families* and *variety evasive subspace families*.

Definition 1.2 (Subspace family). *For* $0 \le k \le n$*, a finite collection (i.e., multiset) of* k*-subspaces of* $\mathbb{P}^n$ *is called a* (projective) k-subspace family *on* $\mathbb{P}^n$*. Similarly, a finite collection of affine* k*-subspaces of* $\mathbb{A}^n$ *is called an* affine k-subspace family *on* $\mathbb{A}^n$*.*

Definition 1.3 (Variety evasive subspace family). *Let* $\mathcal{F}$ *be a family of subvarieties of* $\mathbb{P}^n$ *(resp.* $\mathbb{A}^n$*). Let* $\mathcal{H}$ *be a* k*-subspace family on* $\mathbb{P}^n$ *(resp. affine* k*-subspace family on* $\mathbb{A}^n$*) where* $0 \le k \le n$*. Then:*

- *We say* $\mathcal{H}$ *is* $\mathcal{F}$-evasive *if for every* $\mathcal{V} \in \mathcal{F}$*, there exists* $W \in \mathcal{H}$ *that evades* $\mathcal{V}$*.*
- *We say* $\mathcal{H}$ *is* $(\mathcal{F}, \varepsilon)$-evasive *if for every* $\mathcal{V} \in \mathcal{F}$*, a random element* $W \in \mathcal{H}$ *evades* $\mathcal{V}$ *with probability at least* $1 - \varepsilon$*.*

Connection with hitting sets. Definition 1.3 naturally generalizes the notions of hitting sets in the context of PIT. For example, a collection of points in $\mathbb{P}^n$ is a hitting set for a family $\mathcal{F}$ of homogeneous polynomials in $\mathbb{F}[X_1, \ldots, X_{n+1}]$ iff it is an $\mathcal{F}'$-evasive 0-subspace family, where $\mathcal{F}' = \{\mathcal{V}(P) : P \in \mathcal{F}\}$ is the family of hypersurfaces defined by the polynomials in $\mathcal{F}$. In other words, hitting sets may be viewed as 0-subspace families that are evasive for varieties of codimension one.

Connection with lossless rank condensers. Other than the case of codimension one, we may also consider the special case of

degree one, and this leads to another important family of pseudorandom objects, called *(weak) lossless rank condensers* (Forbes & Guruswami 2015; Forbes *et al.* 2014; Forbes & Shpilka 2012; Gabizon & Raz 2008). These objects were used by Gabizon & Raz (2008) to construct affine extractors. They also play a crucial role in polynomial identity testing (Forbes *et al.* 2014; Forbes & Shpilka 2012; Karnin & Shpilka 2011; Saxena & Seshadhri 2012).

A lossless rank condenser is defined as follows: Let $r \leq t \leq n$ be positive integers. A finite collection $\mathcal{H}$ of matrices $E \in \mathbb{F}^{t\times n}$ is called an (r, L)-*lossless rank condenser* if for every matrix $M \in \mathbb{F}^{n\times r}$ of rank r, the number of $E \in \mathcal{H}$ satisfying $\operatorname{rank}(EM) < r$ is at most L.

The connection between lossless rank condensers and variety evasive subspace families can be seen as follows: Let us assume every matrix $E \in \mathcal{H}$ has full rank t. Such a matrix E corresponds a linear t-subspace W of $\mathbb{F}^n$. On the other hand, a matrix $M \subseteq \mathbb{F}^{n\times r}$ of rank r corresponds to a linear $(n-r)$-subspaces of $\mathbb{F}^n$ via $M \mapsto \ker(M)$, where $\ker(M) = \{u \in \mathbb{F}^n : uM = 0\}$ denotes the left kernel of M. It is easy to see that the condition $\operatorname{rank}(EM) = r$ is equivalent to $\dim(W \cap \ker(M)) = t - r$. Passing from $\mathbb{F}^n$ to $\mathbb{P}^{n-1}$ by taking the quotient modulo scalars, this condition is also equivalent to the condition that the two projective subspaces $\mathbb{P}W$ and $\mathbb{P}(\ker(M))$ evade each other.

Every projective $(n-r-1)$-subspace of $\mathbb{P}^{n-1}$ can be realized as $\mathbb{P}(\ker(M))$ for some rank-r matrix M. Therefore, $\mathcal{H}$ is an (r, L)-lossless rank condenser iff it is an $(\mathcal{F}, \varepsilon)$-evasive $(t-1)$-subspace family on $\mathbb{P}^{n-1}$, where $\varepsilon = L/|\mathcal{H}|$ and $\mathcal{F}$ is the family of all $(n-r-1)$-subspaces of $\mathbb{P}^{n-1}$.

Rank condensers are central objects in the theory of "linear-algebraic pseudorandomness" coined by Forbes & Guruswami (2015). Our study of variety evasive subspace families may be seen as one step of extending the theory to a nonlinear setting.

Explicit lossless rank condensers were used to construct explicit (deterministic) *affine extractors* (Gabizon & Raz 2008) and more generally, *extractors for varieties* (Dvir 2012). Similar ideas were used to construct explicit *deterministic extractors* (and *rank extractors*) for *polynomial sources* (Dvir *et al.* 2009), which also gen-

eralize affine extractors. It is an interesting question to us whether explicit variety evasive subspace families and the related derandomized Noether's normalization lemma (see below) can be similarly useful in this area.

1.2. Our Results. We have seen that variety evasive subspace families generalize some important and well-studied pseudorandom objects. This leads to the following natural question: For which interesting families $\mathcal{F}$ of subvarieties can we construct explicit $\mathcal{F}$-evasive or $(\mathcal{F}, \varepsilon)$-evasive subspace families?

In this paper, we focus on the families of subvarieties of *bounded degree*. First, we recall the definition of the *degree* of a variety.

Definition 1.4 (degree). *The degree of an irreducible variety $\mathcal{V}$ in $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) is the number of intersections of $\mathcal{V}$ with a general projective (resp. affine) subspace of codimension $\dim(\mathcal{V})$. Following Heintz (1983); Heintz & Schnorr (1980), we define the degree of a (possibly reducible) variety to be the sum of the degrees of its irreducible components.*

For convenience, we introduce the following definition.

Definition 1.5. *We say a projective (resp. affine) k-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) is* (n, d)-evasive *if it is $\mathcal{F}$-evasive, where $\mathcal{F}$ is chosen to be the family of all subvarieties of $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) of degree at most d. Similarly, we say $\mathcal{H}$ is* (n, d, ε)-evasive *if it is $(\mathcal{F}, \varepsilon)$-evasive.*

Remark 1.6. *In Definition 1.5, we do not make any assumption about the dimension of the varieties in $\mathcal{F}$ or their irreducible components. We will see in Section 3.1 that in fact, it suffices to consider the subfamily of equidimensional varieties or even irreducible varieties of dimension $n - k - 1$ when constructing variety evasive k-subspace families.*

For $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$, define $N(k, d, n)$ by

$$N(k,d,n) := \min\left\{ \binom{(k'+1)(n'+1+d)}{(k'+1)d}, \binom{(n-k)(n'+1+d)}{(n-k)d} \right\}.$$

where $k' := \min\{k, d-2\} \leq k$ and $n' := k' + n - k \leq n$.

Our main theorem then states as follows.

THEOREM 1.7 (Main Theorem). *For $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\varepsilon \in (0,1)$, there exists an (n, d, ε)-evasive k-subspace family (resp. affine k-subspace family) $\mathcal{H}$ on $\mathbb{P}^n$ (resp. $\mathbb{A}^n$) of size* $\mathrm{poly}(N(k,d,n), n, 1/\varepsilon) \leq \mathrm{poly}(n^{\min\{k+1, n-k, d\}d}, d, 1/\varepsilon)$. *Moreover:*

- *If $\mathrm{char}(\mathbb{F}) = 0$, where $\mathbb{F}$ denotes the base field, then the linear equations defining the projective or affine subspaces in $\mathcal{H}$ are defined over $\mathbb{Q}$. Moreover, the bit-lengths of the numerators and denominators of the coefficients of these linear equations are polynomial in $|\mathcal{H}|$, and the total bit complexity of computing these linear equations is polynomial in $|\mathcal{H}|$.*
- *If $\mathrm{char}(\mathbb{F}) = p > 0$, then the linear equations defining the projective or affine subspaces in $\mathcal{H}$ are defined over an extension field $\mathbb{F}_q$ of $\mathbb{F}_p$, where $q \leq \mathrm{poly}(|\mathcal{H}|, p)$. The total bit complexity of computing these linear equations and constructing the field $\mathbb{F}_q$ is polynomial in $|\mathcal{H}|$ and $\log p$.*

In particular, when d is bounded, the bit complexity of constructing $\mathcal{H}$ is polynomial in n/ε (and $\log p$ if $\mathrm{char}(\mathbb{F}) = p > 0$).

REMARK 1.8. *The two items in the above theorem bound the complexity of the coefficients that define $\mathcal{H}$. The same bounds apply to the coefficients in all constructions presented in this paper, and in particular, to those in Theorem 1.12 and Theorem 1.13 below. These bounds are needed for bounding the bit complexity of the construction of $\mathcal{H}$, which is crucial for demonstrating the explicitness of $\mathcal{H}$. We also remark that if we do not impose any restrictions on the complexity of the coefficients, then it is, in fact, straightforward to construct hitting sets of polynomial size unconditionally (Heintz & Schnorr 1980, Lemma 4.2). This explains why*

we consider bit complexity as the complexity measure rather than assuming that each field operation takes unit cost, which is common in arithmetic complexity.

REMARK 1.9. *A previous version of this paper Guo (2021) proved a weaker upper bound where n' in the definition of $N(k,d,n)$ is replaced by n. Our new bound in Theorem 1.7 has the advantage that when $n-k$ is small, we can get a subspace family of size $\mathrm{poly}(n,1/\varepsilon)$ even if d grows slowly in n:*

- *As $N(k,d,n) \le \binom{(k'+1)(n'+1+d)}{(k'+1)d} \le \binom{(d-1)(n-k+2d-1)}{(d-1)d}$, we can afford any $d \le f(n)$ for some $f(n) = \omega_n(1)$ when $n-k = n^{o(1)}$.*
- *Similarly, as $N(k,d,n) \le \binom{(n-k)(n'+1+d)}{(n-k)d} \le \binom{(n-k)(n-k+2d-1)}{(n-k)d}$, we can afford any $d = O(\log n)$ when $n-k = O(1)$.*

Lower bound. As a complement of the above result, we establish the following *tight* lower bound for projective k-subspace families. It implies that when $n-k = n^{\Omega(1)}$, the assumption of d being bounded is necessary for a projective (n,d)-evasive k-subspace family to have polynomial size.

THEOREM 1.10. *Let $n, d \in \mathbb{N}^+$ and $k \in \{0,1,\dots,n-1\}$. Let $\mathcal{F}$ be the family of equidimensional projective subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. Suppose $\mathcal{H}$ is an $\mathcal{F}$-evasive k-subspace family on $\mathbb{P}^n$. Then*

$$|\mathcal{H}| \ge \begin{cases} (n-k)(k+1)+1 & \text{if } d=1, \\ \max\left\{ d(n-k)(k+1)+1, \binom{d+n-k}{d} + (n-k+1)k \right\} & \text{if } d>1. \end{cases}$$

In particular, $|\mathcal{H}|$ is superpolynomial in n when $n-k = n^{\Omega(1)}$ and $d = \omega(1)$.

Remark 1.11 (Tightness of the lower bound). *When $d = 1$, the lower bound $|\mathcal{H}| \geq (n-k)(k+1)+1$ in Theorem 1.10 is achieved by known explicit lossless rank condensers (Forbes 2014; Forbes et al. 2014; Forbes & Shpilka 2012) (see Section 2.2). For general d, the lower bound in Theorem 1.10 is also tight and matched by* non-explicit *constructions. See Section 4 for a discussion.*

In general, there is a gap between known upper bounds from explicit constructions and the tight lower bound. In particular, when $d \leq (n-k)^{1-\delta}$ for some constant $\delta > 0$, our lower bound gives $|\mathcal{H}| \geq (n-k)^{\Omega(d)} + \mathrm{poly}(n)$ while the upper bound in Theorem 1.7 gives $|\mathcal{H}| \leq (n-k)^{O(d \min\{k,d\})} + \mathrm{poly}(n)$.

Next, we list two applications of our Main theorem (Theorem 1.7): derandomizing Noether's normalization lemma for varieties of low degree, and polynomial identity testing for a special family of depth-4 arithmetic circuits.

1.2.1. Derandomizing Noether's Normalization Lemma. *Noether's normalization lemma*, introduced by Noether (1926), is an important result in commutative algebra and algebraic geometry with many applications. For example, it is used in the development of dimension theory and can be used to prove Grothendieck's generic freeness lemma (Eisenbud 1995). It also has applications in computational algebraic geometry, e.g., computing the dimension of a projective variety (Giusti *et al.* 2000; Giusti & Heintz 1993).

The usual geometric formulation of Noether's normalization lemma states that for any affine variety $\mathcal{V} \subseteq \mathbb{A}^n$ of dimension r, there exists a surjective *finite morphism* $\pi : \mathcal{V} \to \mathbb{A}^r$. (See Section 2.3 for the definition of finite morphisms.) Moreover, π may be chosen to be the restriction of a linear map $\mathbb{A}^n \to \mathbb{A}^r$.[1] There is also a related projective or graded version of the lemma, which

[1]For simplicity, we assume the base field is algebraically closed and hence infinite. But the lemma and our derandomization are valid as long as the field is large enough, depending on the variety $\mathcal{V}$. We remark that Nagata (1962) proved a version of the normalization lemma that is deterministic and does not require the base field to be sufficiently large, but the morphism he used is highly nonlinear. Moreover, Nagata's normalization lemma crucially relies on the fact that the variety is affine, while the normalization lemma we consider here extends to the projective case.

states that for any projective variety $\mathcal{V}$ of dimension r, there exists a surjective finite morphism $\pi : \mathcal{V} \to \mathbb{P}^r$. A special form of this lemma goes back to Hilbert (1890).

In these versions of Noether's normalization lemma, it can be shown that with high probability, a random linear map yields a valid finite morphism π, where "random" means the coefficients of the linear map are chosen randomly from a sufficiently large finite set $S \subseteq \mathbb{F}$. It is thus a natural question to derandomize the lemma.

Mulmuley (2017) studied a form of Noether's normalization lemma and proved that derandomizing it is equivalent to a strengthened form of the black-box derandomization of PIT. There, the ambient projective space has exponential dimension and the problem is constructing a finite morphism $\pi : \mathcal{V} \to \mathbb{P}^k$ with a *succinct* specification in deterministic polynomial time, where $k = \text{poly}(\dim(\mathcal{V}))$ and $\mathcal{V}$ is an *explicit variety* (Mulmuley 2017). This problem was later shown to be in PSPACE (Forbes & Shpilka 2018; Guo *et al.* 2019). The special case for the ring of matrix invariants under simultaneous conjugation was solved in quasipolynomial time by Forbes & Shpilka (2013).

We consider Noether's normalization lemma in its original context and completely derandomize it for projective/affine varieties of bounded degree. The following two theorems summarize our results.

THEOREM 1.12. *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\varepsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^{n+1} \to \mathbb{A}^{r+1}$ of size $\text{poly}(N(k, d, n), n, 1/\varepsilon)$ such that for every subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of dimension r and degree at most d, all but at most ε-fraction of $\pi \in \mathcal{L}$ induce a surjective finite morphism from $\mathcal{V}$ to $\mathbb{P}^r$.*[2] *Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\text{char}(\mathbb{F}) = p > 0$).*

THEOREM 1.13. *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\varepsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^n \to \mathbb{A}^r$ of size $\text{poly}(N(k, d, n-1), n, 1/\varepsilon)$ such that for every*

[2] Let $N(k, d, n) = 1$ when $r = n$ (i.e., $k = -1$). Similarly, in Theorem 1.13, let $N(k, d, n-1) = 1$ when $r = n$ or $r = 0$ (i.e., $k = -1$ or $k = n - 1$).

subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of dimension r and degree at most d, all but at most ε-fraction of $\pi \in \mathcal{L}$ restrict to a surjective finite morphism from $\mathcal{V}$ to $\mathbb{A}^r$. Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).

Dimension-preserving morphisms vs. finite morphisms. Our construction of finite linear morphisms preserves the dimension of a variety of low degree while reducing the dimension of the ambient space. This generalizes the property of lossless rank condensers. However, for the dimension-preserving property, better constructions are known. For example, it can be shown that most of the linear maps $\mathbb{A}^n \to \mathbb{A}^t$ from a lossless rank condenser $\mathcal{H} \subseteq \mathbb{F}^{t \times n}$ already preserve the dimension of a variety $\mathcal{V} \subseteq \mathbb{A}^n$. The intuition here is that $\mathcal{V}$ can be locally approximated at a nonsingular point $p \in \mathcal{V}$ by its *tangent space* at p. (Note that such a nonsingular point always exists when $\mathcal{V}$ is a nonempty variety.) So any linear map that preserves the dimension of this tangent space also preserves the dimension of $\mathcal{V}$.

On the other hand, the morphisms we construct are *finite morphisms*, which are strictly stronger than morphisms that are dimension-preserving. In particular, a finite morphism π always maps a closed set onto a closed set in the Zariski topology. Moreover, the preimage $\pi^{-1}(p)$ of *every* point p in the image of π is a finite set. Neither of these two properties is necessarily satisfied by morphisms that are only dimension-preserving.

These properties of finite morphisms may be useful in extractor theory or other areas. For example, in Theorem 1.13, the cardinality of $\pi^{-1}(p)$ is bounded by the degree of $\mathcal{V}$ for every $p \in \pi(\mathcal{V})$, which translates into a lower bound for the min-entropy of the output of π when the input random source is distributed over the variety $\mathcal{V}$.

1.2.2. Depth-4 Polynomial Identity Testing. Depth-4 arithmetic circuits, also known as $\Sigma\Pi\Sigma\Pi$ circuits, play a very important role in polynomial identity testing. In a surprising result, Agrawal & Vinay (2008) proved that a complete derandomization of black-box PIT for depth-4 circuits implies an $n^{O(\log n)}$-time derandomization of PIT for general circuits of $\mathrm{poly}(n)$ degree.

Dvir & Shpilka (2007) initialized the approach of applying *Sylvester–Gallai* type theorems in geometry to PIT for depth-3 ($\Sigma\Pi\Sigma$) circuits. Extending this approach, Gupta (2014) formulated a conjecture of Sylvester–Gallai type and proved that his conjecture implies a complete derandomization of black-box PIT for depth-4 circuits with bounded top fan-in and bottom fan-in (also called $\Sigma\Pi\Sigma\Pi(k,r)$ circuits, where $k, r = O(1)$). Based on Shpilka (2020) and Peleg & Shpilka (2022), Peleg & Shpilka (2021) proved that this conjecture holds for $k = 3$ and $r = 2$ and used it to give a polynomial time black box PIT algorithm for $\Sigma\Pi\Sigma\Pi(3,2)$ circuits. Using a different approach, Dutta *et al.* (2021) gave a quasipolynomial time black box PIT algorithm for $\Sigma\Pi\Sigma\Pi(k,r)$ circuits. Finally, in an exciting breakthrough, Limaye *et al.* (2024) obtained a subexpoential time black box PIT algorithm for all arithmetic circuits of bounded depth.

Gupta (2014) divided $\Sigma\Pi\Sigma\Pi(k,r)$ into two families: those in a certain Sylvester–Gallai configuration and those that are not. His conjecture states that the circuits in the first family always have bounded *transcendence degree*, depending only on k and r. If the conjecture is true, then the results in Agrawal *et al.* (2016); Beecken *et al.* (2013) imply a complete derandomization of the black-box PIT for this family. For the second family of circuits, which we call *non-SG* circuits, he proved that the black-box PIT can also be derandomized completely.

THEOREM 1.14 (Gupta 2014). *There exists a deterministic black-box PIT algorithm with time complexity* $(dnk)^{\mathrm{poly}(r^{k^2}+k)}$ *for non-SG* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuits of degree at most* d *in* $X_1, \ldots, X_n$ *over* $\mathbb{C}$. *In particular, the algorithm runs in polynomial time when* k *and* r *are bounded.*

Gupta's proof of Theorem 1.14 is quite complex and used tools from computational algebraic geometry, including an effective version of Bertini irreducibility theorem (Heintz & Sieveking 1981) and radical membership testing (which in turn depends on *effective Nullstellensatz* (Dubé 1993; Kollár 1988)).

We observe that what is needed here is simply an explicit construction of subspaces intersecting certain varieties with (at most)

the expected dimension. Plugging in our explicit construction of variety evasive subspace families, we obtain an improved black-box PIT algorithm with a simple proof.

THEOREM 1.15. *There exists a deterministic black-box PIT algorithm with time complexity polynomial in* $d \cdot \binom{k(n+1+r^k)}{kr^k} \cdot \binom{k-1+d}{k-1} \leq \mathrm{poly}(d^k, n^{r^k}, r^{k^2 r^k})$ *(and* $\log p$*, if* $\mathrm{char}(\mathbb{F}) = p > 0$*) for non-SG* $\Sigma\Pi\Sigma\Pi(k, r)$ *circuits of degree at most* d *in* $X_1, \ldots, X_n$ *over an algebraically closed field* $\mathbb{F}$*.*

In particular, Theorem 1.15 improves the exponent of n in the time complexity from $\mathrm{poly}(r^{k^2} + k)$ to $O(r^k)$, and the exponent of d from $\mathrm{poly}(r^{k^2} + k)$ to $O(k)$. Moreover, our proof is more direct and conceptually simpler than the proof in Gupta (2014).

REMARK 1.16. *Mukhopadhyay (2016) gave a deterministic polynomial-time black-box PIT algorithm for* $\Sigma\Pi\Sigma\Pi(k, r)$ *circuits satisfying a variant of the non-SG assumption. (Its time complexity is similar to the time complexity in Theorem 1.14.) It appears to us that his assumption in fact implies the non-SG assumption. The main tool used there is the* multivariate resultant*, which may be related to our approach based on Chow forms (see Section 1.3). Indeed, it is known that a multivariate resultant is the Chow form of a Veronese variety (Gelfand et al. 1994, Chapter 3, Example 2.4).*

1.3. Proof Overview. We present an overview of our proof of Theorem 1.7 and that of Theorem 1.10.

Overview of the proof of Theorem 1.7. In the proof of Theorem 1.7, we focus on constructing a k-subspace family on $\mathbb{P}^n$. The case of $\mathbb{A}^n$ can be easily derived from it by viewing $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$ and restricting to this subset.

Consider a variety $\mathcal{V} \subseteq \mathbb{P}^n$ of degree at most d. We want to construct a k-subspace family $\mathcal{H}$ on $\mathbb{P}^n$, independent of $\mathcal{V}$, such that all but at most ε-fraction of $W \in \mathcal{H}$ evade $\mathcal{V}$. Our key ideas can be summarized as follows.

Reducing to the equidimensional/irreducible case of dimension $n-k-1$. As a first step, we reduce the problem to the special case that $\mathcal{V}$ is an *equidimensional* (or even irreducible) variety of $\mathbb{P}^n$ of dimension $n-k-1$, which means every irreducible component of $\mathcal{V}$ has dimension exactly $n-k-1$. This step is explained in Section 3.1.

Hitting the Chow form of $\mathcal{V}$. Denote by $\mathbb{G}(k,n)$ the Grassmannian consisting of of all k-subspaces of $\mathbb{P}^n$. As $\operatorname{codim}(\mathcal{V}) = n-(n-k-1) > k$, a *general* k-subspace $W \in \mathbb{G}(k,n)$ is disjoint from $\mathcal{V}$, but we want to find such W explicitly.

One remarkable fact in algebraic geometry is that there is a single polynomial $\widetilde{R}_{\mathcal{V}}$ on the Grassmannian $\mathbb{G}(k,n)$ that defines precisely the subset of k-subspaces that intersect $\mathcal{V}$. This polynomial $\widetilde{R}_{\mathcal{V}}$ is called the *Chow form* of $\mathcal{V}$ (in *Stiefel coordinates*). Chow forms are also known as *associated forms*, *Cayley forms*, or *Cayley–van der Waerden–Chow forms* in literature. They were introduced by Cayley (1860) to represent curves in $\mathbb{P}^3$ and later generalized by Chow & van der Waerden (1937). See Dalbec & Sturmfels (1995) for an introduction to Chow forms and Gelfand *et al.* (1994) for an exposition in the context of elimination theory.

To be more specific, for a k-subspace $W \in \mathbb{G}(k,n)$, we choose a $(k+1)\times(n+1)$ matrix A that represents W. The Chow form $\widetilde{R}_{\mathcal{V}}$ is a polynomial of degree $(k+1)\deg(\mathcal{V})$ in $(k+1)(n+1)$ variables with the following property: $\widetilde{R}_{\mathcal{V}}$ vanishes at the matrix A (viewed as a list of $(k+1)(n+1)$ coordinates) if and only if $\mathcal{V}\cap W \neq \emptyset$. Thus, $\widetilde{R}_{\mathcal{V}}$ defines precisely the subset of "bad" k-subspaces that we want to avoid.

Therefore, the problem becomes finding a collection of $(k+1)\times(n+1)$ matrices of full rank that "hit" the polynomial $\widetilde{R}_{\mathcal{V}}$ of degree $(k+1)\deg(\mathcal{V}) \le (k+1)d$. Using black-box PIT for low degree polynomials (see Section 2.1), we are able to construct an (n,d,ε)-evasive k-subspace family of size polynomial in $\binom{(k+1)(n+1+d)}{(k+1)d}$ and $1/\varepsilon$, which is $\operatorname{poly}(n,1/\varepsilon)$ when k and d are both bounded. A similar "dual" construction yields a k-subspace family of size polynomial in $\binom{(n-k)(n+1+d)}{(n-k)d}$ and $1/\varepsilon$, which is $\operatorname{poly}(n,1/\varepsilon)$ when both $n-k$ and d are bounded. For applications where d is small and

either k or $n-k$ is small (e.g., Theorem 1.15), these constructions are good enough. However, when k and $n-k$ are both linear in n, the resulting k-subspace families have exponential size in n, even if d is bounded.

A two-step construction. To obtain a good construction for *arbitrary* dimension k, we use a connection with Noether normalization. It is a standard fact in algebraic geometry that a subspace $W \subseteq \mathbb{P}^n$ is disjoint from a variety $\mathcal{V}$ iff it defines a *projection* $\mathbb{P}^n \setminus W \to \mathbb{P}^{n-\dim(W)-1}$ that restricts to a finite morphism from $\mathcal{V}$ to $\mathbb{P}^{n-\dim(W)-1}$. Thus, we may reformulate our problem as finding finite morphisms $\pi : \mathcal{V} \to \mathbb{P}^{n-k-1}$ that come from projections.

We also need another fact, which states that the codimension of an irreducible subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ in $\mathrm{span}(\mathcal{V})$ is at most $\deg(\mathcal{V})-1$, where $\mathrm{span}(\mathcal{V})$ denotes the smallest projective subspace containing $\mathcal{V}$ (see Lemma 3.13). Therefore, for irreducible $\mathcal{V}$ of degree at most d, there exists a projective subspace Λ of dimension (at most) $\dim(\mathcal{V})+d-1$ that contains $\mathcal{V}$.

Our idea is to use a two-step construction. Namely, we first construct a finite morphism $\pi_1 : \mathcal{V} \to \mathbb{P}^{\dim(\Lambda)}$ that comes from a projection, and then construct another finite morphism $\pi_2 : \mathcal{V}' \to \mathbb{P}^{n-k-1}$, where $\mathcal{V}' := \pi_1(\mathcal{V}) \subseteq \mathbb{P}^{\dim(\Lambda)}$. Composing π_1 with π_2 yields a desired map from $\mathcal{V}$ to $\mathbb{P}^{n-k-1}$.

The first step is just the problem of constructing lossless rank condensers, which has an optimal solution (Forbes *et al.* 2014; Forbes & Shpilka 2012) (see Section 2.2). For the second step, we need to hit the Chow form of $\mathcal{V}'$. Thanks to the first step, the codimension of $\mathcal{V}'$ in $\mathbb{P}^{\dim(\Lambda)}$ is only $\dim(\Lambda)-\dim(\mathcal{V}') = \dim(\Lambda)-\dim(\mathcal{V}) \le d-1$. As the codimension is low, we may use black-box PIT for low degree polynomials just like before, and Theorem 1.7 follows.

Finally, the above connection with Noether normalization also allows us to derive Theorem 1.12 and Theorem 1.13 easily from Theorem 1.7.

Overview of the proof of Theorem 1.10. Our lower bound (Theorem 1.10) follows from a dimension counting argument. Let

$C(r, d, n)$ be the set of all varieties $\mathcal{V} \subseteq \mathbb{P}^n$ of dimension $r := n - k - 1$ and degree d, which is the set of varieties that we want to evade.

Roughly speaking, the idea is to show that (1) $C(r, d, n)$ itself can be realized as a subvariety of some projective space $\mathbb{P}^N$, and (2) for every k-subspace W, the subset of $\mathcal{V} \in C(r, d, n)$ that W fails to evade is the intersection of $C(r, d, n)$ with some hyperplane H_W of $\mathbb{P}^N$.

To see how (1) and (2) above lead to a lower bound, suppose $\mathcal{H}$ is a $C(r, d, n)$-evasive k-subspace family, i.e., for any $\mathcal{V} \in C(r, d, n)$, there exists $W \in \mathcal{H}$ that is disjoint from $\mathcal{V}$. Then the intersection $C(r, d, n) \cap \bigcap_{W \in \mathcal{H}} H_W$ must be empty. On the other hand, taking the intersection with each hyperplane H_W reduces the dimension of a projective variety by at most one. So we have a lower bound $|\mathcal{H}| \geq \dim(C(r, d, n)) + 1$.

How do we realize $C(r, d, n)$ as a subvariety of $\mathbb{P}^N$? It turns out that this is a classical problem in the study of moduli spaces and a solution was given by Cayley (1860) and Chow & van der Waerden (1937) using the *Chow embedding*: The Chow embedding $C(r, d, n) \to \mathbb{P}^N$ simply sends a variety $\mathcal{V}$ to its Chow form $\widetilde{R}_{\mathcal{V}}$, where $\widetilde{R}_{\mathcal{V}}$ is viewed as a point in the projective space $\mathbb{P}^N$ whose homogeneous coordinates are given by the coefficients of $\widetilde{R}_{\mathcal{V}}$. (The actual Chow embedding we use has a slightly different form, which is essentially equivalent to the one described here.)

A technical issue here is that the image of $C(r, d, n)$ under the Chow embedding is generally not closed in the Zariski topology. To fix this issue, the definition of $C(r, d, n)$ needs to be modified so that it contains not only subvarieties of $\mathbb{P}^n$, but also *(effective) algebraic cycles* on $\mathbb{P}^n$, which are a generalization of subvarieties. A theorem of Chow & van der Waerden (1937) then states that the Chow embedding does embed $C(r, d, n)$ in a projective subspace $\mathbb{P}^N$ as a subvariety, known as a *Chow variety*.

Finally, we also need a lower bound for the dimension of the Chow variety $C(r, d, n)$. In fact, the exact value of $\dim(C(r, d, n))$ was determined by Azcue (1992) and independently by Lehmann (2017). Plugging in the value of $\dim(C(r, d, n))$ proves Theorem 1.10.

1.4. Other Related Work. Dvir *et al.* (2014) constructed explicit *variety evasive sets*, which are large subsets of $\mathbb{F}_q^n$ over a finite field $\mathbb{F}_q$ that have small intersection with affine varieties of fixed dimension and bounded degree. It generalizes an earlier construction of *subspace evasive sets* of Dvir & Lovett (2012). The definition of evasiveness there is different from ours, but they are related, since a key step in the proofs of Dvir *et al.* (2014); Dvir & Lovett (2012) is proving the intersection of two varieties has dimension zero. We also note that a subspace/variety evasive set is a single set, defined in a highly nonlinear way, whereas we define a variety evasive subspace family to be a collection of projective or affine subspaces. Finally, the results in Dvir *et al.* (2014); Dvir & Lovett (2012) hold only for affine subspaces/subvarieties, whereas we give our construction first in the projective setting and then derive the affine counterpart from it.

Guruswami & Xing (2013) introduced a related notion called *subspace designs*. A subspace design is a collection $\mathcal{H}$ of large subspaces of $\mathbb{F}^n$ such that for any small subspace $V \subseteq \mathbb{F}^n$, the number of $W \in \mathcal{H}$ satisfying $\dim(W \cap V) > 0$ is small (or even the sum $\sum_{W \in \mathcal{H}} \dim(W \cap V)$ is small). An equivalence between subspace designs and lossless rank condensers was proved in Forbes & Guruswami (2015). Explicit subspace designs were constructed by Guruswami & Kopparty (2016) and also by Guruswami *et al.* (2018). They have applications to constructing explicit list-decodable codes with small list size (Guo & Ron-Zewi 2021; Guruswami *et al.* 2016; Guruswami & Xing 2013; Kopparty *et al.* 2023) and explicit dimension expanders (Forbes & Guruswami 2015; Guruswami *et al.* 2021). Subspace designs were also used to prove lower bounds in communication complexity (Chattopadhyay *et al.* 2021).

Jeronimo *et al.* (2004) gave a randomized algorithm, in the Blum–Shub–Smale model over fields of characteristic zero, that computes the Chow forms of varieties defined by input polynomials. The (expected) time complexity of their algorithm is polynomial in the sizes of the arithmetic circuits encoding the input polynomials and the *geometric degree* of the polynomial system. See also the survey by Krick (2002).

Chow varieties of effective zero-cycles and their higher secant varieties are related to lower bounds for depth-3 arithmetic circuits. They have received a considerable amount of attention in Geometric Complexity Theory (Landsberg 2012, 2015).

Organization of the paper. Preliminaries and notations are given in Section 2. We prove Theorem 1.7, Theorem 1.12, and Theorem 1.13 in Section 3. In Section 4, we prove the lower bound (Theorem 1.10) and also give a non-explicit construction that matches this lower bound. The application to PIT for depth-4 circuits (Theorem 1.15) is explained in Section 5. Finally, we list some open problems and future directions in Section 6.

2. Preliminaries and Notations

Define $\mathbb{N} := \{0, 1, 2 \dots\}$ and $\mathbb{N}^+ := \{1, 2, \dots\}$. Let $[n] := \{1, 2, \dots, n\}$ for $n \in \mathbb{N}$. For a set S and $k \in \mathbb{N}$, denote by $\binom{S}{k}$ the set of all subsets of S of cardinality k.

Denote by $\mathbb{F}$ an algebraically closed field throughout this paper. We use notations like $\mathbb{F}[X_{i,j} : i \in [n], j \in [m]]$ to denote the polynomial ring over $\mathbb{F}$ in a finite set of variables (in this case, in the set of variables $\{X_{i,j} : i \in [n], j \in [m]\}$). The vector space of $n \times m$ matrices over $\mathbb{F}$ is denoted by $\mathbb{F}^{n \times m}$.

For an $n \times m$ matrix A and subsets $S \subseteq [n]$, $T \subseteq [m]$, denote by $A_{S,T}$ the submatrix of A whose rows and columns are selected by S and T respectively, where the orderings of rows and columns are preserved.

2.1. Black-Box PIT for Low Degree Polynomials.

For convenience, we strengthen the definition of hitting sets as follows.

Definition 2.1 (ε-hitting set). *Let $\mathcal{F}$ be a family of polynomials in $\mathbb{F}[X_1, \dots, X_n]$ and $\varepsilon \in (0, 1)$. We say a finite collections of points $\mathcal{H} \subseteq \mathbb{F}^n$ is an ε-*hitting set *for $\mathcal{F}$ if for any nonzero $Q \in \mathcal{F}$, the evaluation $Q(\alpha)$ is nonzero for all but at most ε-fraction of $\alpha \in \mathcal{H}$.*

We need an explicit construction of ε-hitting sets for low degree polynomials. This problem has been well studied (Bläser & Pandey

2020; Bogdanov 2005; Bshouty 2014; Cohen & Ta-Shma 2013; Demillo & Lipton 1978; Klivans & Spielman 2001; Lu 2012; Schwartz 1980; Zippel 1979). For completeness, we present a construction based on sparse polynomial identity testing.

Recall that a polynomial is *s-sparse* if it has at most s monomials. We need the following lemma from Agrawal *et al.* (2015).

LEMMA 2.2 (Agrawal *et al.* 2015, Lemma 4, restated). *For* $n, s, d \in \mathbb{N}^+$ *and* $\varepsilon_0 \in (0,1)$, *there exist maps* $w_1, w_2, \ldots, w_N : [n] \to [N \log N]$, *where* $N = \text{poly}(n, s, \log d, \varepsilon_0^{-1})$, *such that for any nonzero s-sparse polynomial* $f \in \mathbb{F}[X_1, \ldots, X_n]$ *of individual degree at most* d, *all but at most* ε_0*-fraction of* w_i *among* $w_1, w_2, \ldots, w_N$ *satisfies* $f(Y^{w_i(1)}, \ldots, Y^{w_i(n)}) \neq 0$. *Moreover, the bit complexity of computing* $w_1, w_2, \ldots, w_N$ *is polynomial in* N.

Given $n, d \in \mathbb{N}^+$ and $\varepsilon \in (0,1)$, we construct an ε-hitting set for n-variate polynomials of degree at most d as follows:

1. Let $s = \binom{n+d}{d}$, $\varepsilon_0 = \varepsilon/2$, and $M = \lceil \varepsilon_0^{-1} dN \log N \rceil$, where N is as in Lemma 2.2.

2. Let $w_1, \ldots, w_N$ be as in Lemma 2.2, which can be computed in time $\text{poly}(N)$.

3. If $\text{char}(\mathbb{F}) = 0$, let $S = [M] \subseteq \mathbb{Z} \subseteq \mathbb{F}$. If $\text{char}(\mathbb{F}) = p > 0$, choose the smallest p-power q such that $q \geq M$, and choose S to be a subset of $\mathbb{F}_q \subseteq \mathbb{F}$ of cardinality M. We remark that $\mathbb{F}_q$ can be constructed deterministically in time $\text{poly}(M, \log p)$. To see this, note that $q \leq Mp$ by the minimality of q. If $M \leq p$, then $\mathbb{F}_q$ is just $\mathbb{F}_p$. On the other hand, if $p < M \leq q$, then $\mathbb{F}_q$ can be constructed in time $\text{poly}(p, [\mathbb{F}_q : \mathbb{F}_p])$ (see, e.g., Lenstra (1990)), which is polynomial in M since $p < M$ and $q \leq Mp$.

4. Finally, construct the following collection of points in $\mathbb{F}^n$ of size MN

$$T = \{(\alpha^{w_i(1)}, \ldots, \alpha^{w_i(n)}) : \alpha \in S, i \in [N]\} \subseteq \mathbb{F}^n.$$

LEMMA 2.3. *For any nonzero polynomial $f \in \mathbb{F}[X_1, \ldots, X_n]$ of degree at most d, we have $f(u) \neq 0$ for all but at most ε-fraction of $u \in T$. The collection T has cardinality* $\mathrm{poly}\left(\binom{n+d}{d}, 1/\varepsilon\right)$ *and can be computed in time* $\mathrm{poly}(|T|)$.

PROOF. Let $f \in \mathbb{F}[X_1, \ldots, X_n]$ be a nonzero polynomial of degree at most d. Note that f is trivially s-sparse, where $s = \binom{n+d}{d}$. So by Lemma 2.2, for all but at most ε_0-fraction of $i \in [N]$, we have $\widetilde{f}_i := f(Y^{w_i(1)}, \ldots, Y^{w_i(n)}) \neq 0$. Consider $i \in [N]$ such that $\widetilde{f}_i \neq 0$. Note that $\widetilde{f}_i$ is a univariate polynomial of degree at most $dN \log N$. So it has at most $dN \log N \leq \varepsilon_0 M$ zeros. Therefore, by the choice of M, we have $f(\alpha^{w_i(1)}, \ldots, \alpha^{w_i(n)}) = \widetilde{f}_i(\alpha) \neq 0$ for all but at most ε_0-fraction of $\alpha \in S$. It follows that $f(u) \neq 0$ holds for all but at most ε-fraction of $u \in T$, as claimed. The rest of the lemma follows easily from the construction. □

Note that the seed length required to choose a random element in T is $\log |T| = O(\log \binom{n+d}{d} + \log(1/\varepsilon))$, which is optimal up to a constant factor. We have made no effort to optimize the constant hidden in $O(\cdot)$. Interested readers may find the state-of-the-art result in Bläser & Pandey (2020), which achieves the optimal constant, at least when $d = n^{o(1)}$.

2.2. Explicit Lossless Rank Condensers. We need the following lemma in the context of *lossless rank condensers*. The construction in the lemma was given by Forbes & Shpilka (2012) and the lemma itself follows implicitly from the analysis of Forbes *et al.* (2014). It was also stated explicitly as Forbes (2014, Theorem 5.4.3).

LEMMA 2.4 (Forbes 2014; Forbes *et al.* 2014). *Let $n \in \mathbb{N}^+$ and $r \in [n]$. Let $\omega \in \mathbb{F}^\times$ such that the multiplicative order of ω is at least n. Define the $r \times n$ matrix $W = (w_{i,j})_{i \in [r], j \in [n]}$ over $\mathbb{F}[X]$ by*

$$w_{i,j} = (\omega^{i-1} X)^{j-1}.$$

Then for every $n \times r$ matrix M over $\mathbb{F}$ of rank r, the polynomial $\det(WM) \in \mathbb{F}[X]$ is nonzero and has degree at most $r(n-r)$ after dividing out powers of X.

COROLLARY 2.5. *Let n, r, W be as in Lemma 2.4 and $\varepsilon \in (0,1)$. Let $S \subseteq \mathbb{F}^\times$ be a finite set of cardinality at least $r(n-r)/\varepsilon$. For every $n \times r$ matrix M over $\mathbb{F}$ of rank r, we have $\mathrm{rank}(W(\alpha)M) = r$ for all but at most ε-fraction of $\alpha \in S$, where $W(\alpha)$ denotes the matrix $(w_{i,j}(\alpha))_{i\in[r],j\in[n]}$ over $\mathbb{F}$.*

Corollary 2.5 states that the collection $\{W(\alpha) : \alpha \in S\}$ of matrices is a (weak) $(r, \varepsilon|S|)$-*lossless rank condenser*, as defined in Forbes & Guruswami (2015). Note that for each $\alpha \in S$, we have $\mathrm{rank}(W(\alpha)) = r$ and hence $W(\alpha)$ corresponds to an $(r-1)$-subspace $U_{W(\alpha)}$ of $\mathbb{P}^{n-1}$. As explained in Introduction, the collection $\mathcal{H} = \{U_{W(\alpha)} : \alpha \in S\}$ is an $(\mathcal{F}, \varepsilon)$-evasive $(r-1)$-subspace family on $\mathbb{P}^{n-1}$, where $\mathcal{F}$ is the family of $(n-r-1)$-subspaces of $\mathbb{P}^{n-1}$. Choosing S of size $r(n-r)+1$ and $\varepsilon = 1 - \frac{1}{r(n-r)+1}$ shows that the lower bound in Theorem 1.10 is achieved when $d = 1$.

2.3. Preliminaries on Algebraic Geometry. We list basic preliminaries and notations on algebraic geometry used in this paper. One can also refer to a standard text, e.g., Harris (1992); Shafarevich (1994).

Affine and projective spaces. For $n \in \mathbb{N}$, write $\mathbb{A}^n$ for the *affine n-space* over $\mathbb{F}$. It is defined to be the set $\mathbb{F}^n$ equipped with the *Zariski topology*, defined as follows: A subset $S \subseteq \mathbb{A}^n$ is *(Zariski-)closed* if it is the set of common zeros of a set of polynomials in $\mathbb{F}[X_1, \ldots, X_n]$. The complement of a closed set is an *open* set. The origin of an affine space is denoted by $\mathbf{0}$.

Write $\mathbb{P}^n$ for the *(projective) n-space* over $\mathbb{F}$, defined to be the quotient set $(\mathbb{A}^{n+1} \setminus \{\mathbf{0}\})/\sim$, where $\sim$ is the equivalence relation defined by scaling, i.e., $u \sim v$ if $u = cv$ for some $c \in \mathbb{F}^\times$. The set $\mathbb{P}^n$ is again equipped with the *Zariski topology*, where a subset is closed if it is the set of common zeros of a set of *homogeneous* polynomials in $\mathbb{F}[X_1, \ldots, X_{n+1}]$. We use $(n+1)$-tuples $(x_1, \ldots, x_{n+1})$ to represent points in $\mathbb{P}^n$, called *homogeneous coordinates*.

For a vector space V over $\mathbb{F}$ of dimension $n+1$, where $n \in \mathbb{N}$, define the projective space $\mathbb{P}V = (V \setminus \{\mathbf{0}\})/\sim$, where $\sim$ is again the equivalence relation defined by scaling. By fixing a coordinate system of V and identifying it with $\mathbb{A}^{n+1}$, we may identify $\mathbb{P}V$ with

$\mathbb{P}^n$.

Varieties. *Varieties* in this paper refer to either projective or affine varieties. A *projective (resp. affine) variety* is simply a closed subset of a projective (resp. affine) subspace. If $\mathcal{V}_1$ and $\mathcal{V}_2$ are closed subsets of a projective or affine space and $\mathcal{V}_1 \subseteq \mathcal{V}_2$, we say $\mathcal{V}_1$ is a *(closed) subvariety* of $\mathcal{V}_2$.

A variety is *reducible* if it is the union of finitely many proper subvarieties, and otherwise *irreducible*. Affine and projective spaces are irreducible. A variety $\mathcal{V}$ can be uniquely written as the union of finitely many maximal irreducible subvarieties, which are called the *irreducible components* of $\mathcal{V}$.

A projective or affine variety is called a *hypersurface* (resp. *hyperplane*) if it is definable by a single polynomial (resp. single linear polynomial).

Hilbert's Nullstellensatz. An ideal I of a commutative ring R is *radical* if $a^m \in I$ implies $a \in I$ for every $a \in R$ and $m \in \mathbb{N}^+$. For an ideal I of $\mathbb{F}[X_1, \ldots, X_n]$, denote by $\mathcal{V}(I)$ the subvariety of $\mathbb{A}^n$ defined by the polynomials in I. Define $\mathcal{V}(f_1, \ldots, f_k) = \mathcal{V}(\langle f_1, \ldots, f_k \rangle)$ for $f_1, \ldots, f_k \in \mathbb{F}[X_1, \ldots, X_n]$. For a subvariety $\mathcal{V}$ of $\mathbb{A}^n$, denote by $I(\mathcal{V})$ the ideal of $\mathbb{F}[X_1, \ldots, X_n]$ consisting of all the polynomials vanishing on $\mathcal{V}$. *Hilbert's Nullstellensatz* states that the map $\mathcal{V} \mapsto I(\mathcal{V})$ is an inclusion-reversing one-to-one correspondence between the subvarieties of $\mathbb{A}^n$ and the radical ideals of $\mathbb{F}[X_1, \ldots, X_n]$, with the inverse map $I \mapsto \mathcal{V}(I)$.

For a subvariety $\mathcal{V}$ of $\mathbb{A}^n$, define $\mathbb{F}[\mathcal{V}] := \mathbb{F}[X_1, \ldots, X_n]/I(\mathcal{V})$, called the *coordinate ring* of $\mathcal{V}$.

Projective Nullstellensatz. Consider the polynomial ring $R = \mathbb{F}[X_1, \ldots, X_{n+1}]$. It can be written as a direct sum $R = \bigoplus_{d=0}^{\infty} R_d$ where each R_d denotes the space of degree-d homogeneous polynomials, called the *homogeneous part of degree d* of R or simply the *degree-d part* of R. For an ideal I of R and $d \in \mathbb{N}$, let $I_d := I \cap R_d$, called the *degree-d part* of I. We say I is a *homogeneous ideal* if $I = \bigoplus_{d=0}^{\infty} I_d$. For a homogeneous ideal I of R, we have $R/I = \bigoplus_{d=0}^{\infty} (R/I)_d$ where $(R/I)_d := R_d/I_d$.

For a homogeneous ideal I of R, denote by $\mathcal{V}(I)$ the subvariety of $\mathbb{P}^n$ defined by the homogeneous polynomials in I. Define $\mathcal{V}(f_1, \ldots, f_k) = \mathcal{V}(\langle f_1, \ldots, f_k \rangle)$ for homogeneous polynomials $f_1, \ldots, f_k \in R$. For a subvariety $\mathcal{V}$ of $\mathbb{P}^n$, denote by $I(\mathcal{V})$ the ideal generated by the homogeneous polynomials vanishing on $\mathcal{V}$, which is a homogeneous ideal. The *projective Nullstellensatz* states that the map $\mathcal{V} \mapsto I(\mathcal{V})$ is an inclusion-reversing one-to-one correspondence between the nonempty subvarieties of $\mathbb{P}^n$ and the radical homogeneous ideals of R properly contained in $\langle X_1, \ldots, X_{n+1} \rangle$, with the inverse map $I \mapsto \mathcal{V}(I)$.

For a subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ and the corresponding homogeneous ideal $I = I(\mathcal{V})$, we say R/I is the *homogeneous coordinate ring* of $\mathcal{V}$.

Morphisms. Let $\mathcal{V}_1 \subseteq \mathbb{A}^n$ and $\mathcal{V}_2 \subseteq \mathbb{A}^m$ be affine varieties. A *morphism* from $\mathcal{V}_1$ to $\mathcal{V}_2$ is a map $f : \mathcal{V}_1 \to \mathcal{V}_2$ that is a restriction of a polynomial map $\mathbb{A}^n \to \mathbb{A}^m$. Such a morphism f is associated with a ring homomorphism $f^\sharp : \mathbb{F}[\mathcal{V}_2] \to \mathbb{F}[\mathcal{V}_1]$, making $\mathbb{F}[\mathcal{V}_1]$ an algebra over $\mathbb{F}[\mathcal{V}_2]$. We say f is *finite* if $\mathbb{F}[\mathcal{V}_1]$ is finitely generated as an $\mathbb{F}[\mathcal{V}_2]$-module.

Let $f : \mathcal{V}_1 \to \mathcal{V}_2$ be a map between projective varieties $\mathcal{V}_1$ and $\mathcal{V}_2$. We say f is a morphism from $\mathcal{V}_1$ to $\mathcal{V}_2$ if there exists a collection of open subsets $\{U_i\}_{i \in I}$ of $\mathcal{V}_2$ such that $\mathcal{V}_2 = \bigcup_{i \in I} U_i$ (i.e., $\{U_i\}_{i \in I}$ is an open cover of $\mathcal{V}_2$) and for each $i \in I$, the restriction $f|_{f^{-1}(U_i)} : f^{-1}(U_i) \to U_i$ is a morphism between affine varieties. Furthermore, if each $f|_{f^{-1}(U_i)}$ is finite, then we say f is finite. Finiteness does not depend on the choice of the affine open cover. Namely, if $f : \mathcal{V}_1 \to \mathcal{V}_2$ is a finite morphism between projective varieties $\mathcal{V}_1$ and $\mathcal{V}_2$, and U is an open subset of $\mathcal{V}_2$ such that $f|_{f^{-1}(U)} : f^{-1}(U) \to U$ is a morphism between affine varieties, then $f|_{f^{-1}(U)}$ is also finite.

The image of a morphism $f : \mathcal{V}_1 \to \mathcal{V}_2$ is denoted by $\mathrm{Im}(f)$ or $f(\mathcal{V}_1)$. The image of a closed set under a finite morphism is still closed. The composition of two finite morphisms is still finite.

Dimension. The *dimension* of an irreducible variety $\mathcal{V}$, denoted by $\dim(\mathcal{V})$, is the largest integer m such that there exists a chain of irreducible varieties $\emptyset \subsetneq \mathcal{V}_0 \subsetneq \mathcal{V}_1 \subsetneq \cdots \subsetneq \mathcal{V}_m = \mathcal{V}$. More generally, the dimension of a nonempty variety is the maximal dimension of its irreducible components. We define the dimension of an empty set to be $-\infty$. A variety is *equidimensional* if its irreducible components have the same dimension.

If $\pi : \mathcal{V} \to \mathcal{V}'$ is a finite morphism, then $\dim(\mathcal{V}) = \dim(\pi(\mathcal{V}))$.

Degree. The *degree* of an irreducible subvariety $\mathcal{V}$ of $\mathbb{P}^n$ (resp. $\mathbb{A}^n$), denoted by $\deg(\mathcal{V})$, is the number of intersections of $\mathcal{V}$ with a projective (resp. affine) subspace of codimension $\dim(\mathcal{V})$ in general position. More generally, we define the degree of a subvariety of $\mathbb{P}^n$ or $\mathbb{A}^n$ to be the sum of the degrees of its irreducible components.

Projective closure. The affine n-space $\mathbb{A}^n$ may be regarded as an open subset of $\mathbb{P}^n$ via the map $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. The complement $H_\infty := \mathbb{P}^n \setminus \mathbb{A}^n$ is a hyperplane of $\mathbb{P}^n$ defined by $X_{n+1} = 0$, called the *hyperplane at infinity*. For an affine subvariety $\mathcal{V}$ of $\mathbb{A}^n \subseteq \mathbb{P}^n$, the smallest projective subvariety of $\mathbb{P}^n$ containing $\mathcal{V}$ is the *projective closure* of $\mathcal{V}$, which we denote by $\mathcal{V}_{\mathrm{cl}}$. It is known that $\mathcal{V}_{\mathrm{cl}} \cap \mathbb{A}^n = \mathcal{V}$, $\dim(\mathcal{V}_{\mathrm{cl}}) = \dim(\mathcal{V})$, and $\deg(\mathcal{V}_{\mathrm{cl}}) = \deg(\mathcal{V})$.

Joins of disjoint projective varieties. For two distinct points $p, q \in \mathbb{P}^n$, denote by $\overline{pq}$ the unique projective line passing through them. For two *disjoint* projective subvarieties $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathbb{P}^n$, define the *join* $J(\mathcal{V}_1, \mathcal{V}_2)$ of $\mathcal{V}_1$ and $\mathcal{V}_2$ as

$$J(\mathcal{V}_1, \mathcal{V}_2) := \bigcup_{p \in \mathcal{V}_1, q \in \mathcal{V}_2} \overline{pq}.$$

LEMMA 2.6 (Harris 1992, Examples 6.17, 11.36, and 18.17). *$J(\mathcal{V}_1, \mathcal{V}_2)$ is a subvariety of $\mathbb{P}^n$ of dimension $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) + 1$ and degree at most $\deg(\mathcal{V}_1) \cdot \deg(\mathcal{V}_2)$.*

We also need the following facts.

LEMMA 2.7 (Harris 1992, Exercise 11.6 and Corollary 18.5). *Let $\mathcal{V}$ be a nonempty equidimensional subvariety of $\mathbb{P}^n$ and H a hypersurface of $\mathbb{P}^n$ not containing an irreducible component of $\mathcal{V}$. Then $\mathcal{V} \cap H$ is an equidimensional subvariety of dimension $\dim(\mathcal{V}) - 1$ and degree at most $\deg(\mathcal{V}) \cdot \deg(H)$ (or an empty set if $\dim(\mathcal{V}) = 0$).*

COROLLARY 2.8. *Let $\mathcal{V}$ be a subvariety of $\mathbb{P}^n$ of dimension r, where $0 \leq r < n$. Then there exists an $(n - r - 1)$-subspace W disjoint from $\mathcal{V}$.*

PROOF. It suffices to show that there exist hyperplanes $H_1, \ldots, H_{r+1}$, such that $\mathcal{V}_i := \mathcal{V} \cap (\bigcap_{j=1}^{i} H_j)$ is empty for some $i \leq r + 1$. We may inductively choose each H_i such that $C \not\subseteq H_i$ for every irreducible component C of $\mathcal{V}_{i-1}$, so that $\dim(\mathcal{V}_i) \leq \dim(\mathcal{V}_{i-1}) - 1$ by Lemma 2.7. So $\mathcal{V}_i = \emptyset$ for some $i \leq r + 1$. □

LEMMA 2.9 (Shafarevich 1994, Section I.6.2, Theorem 6). *Suppose $\mathcal{V}_1$ and $\mathcal{V}_2$ are subvarieties of $\mathbb{P}^n$ and $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_1) \geq n$. Then $\mathcal{V}_1 \cap \mathcal{V}_2 \neq \emptyset$ and $\dim(\mathcal{V}_1 \cap \mathcal{V}_2) \geq \dim(\mathcal{V}_1) + \dim(\mathcal{V}_1) - n$.*

3. Proof of the Main Theorem

In this section, we prove the Main theorem (Theorem 1.7) together with Theorem 1.12 and Theorem 1.13. In Section 3.1, we show that it suffices to consider equidimensional or irreducible subvarieties of dimension $n - k - 1$. Section 3.2 contains an introduction to Chow forms. In Section 3.3, we present the explicit constructions and complete the proof of Theorem 1.7. As a product, Theorem 1.12 and Theorem 1.13 are also proved in Section 3.3.

3.1. Reducing to the Case of Equidimensional or Irreducible Varieties. The following lemma states that to construct k-subspace families that are evasive for subvarieties of $\mathbb{P}^n$, it suffices to consider equidimensional subvarieties of dimension $n-k-1$ (i.e., codimension $k+1$).

LEMMA 3.1. *Let $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$. Let $\mathcal{F}$ be the family of all equidimensional subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. Then an $(\mathcal{F}, \varepsilon)$-evasive k-subspace family is also (n, d, ε)-evasive.*

The proof of Lemma 3.1 is based on the following claim.

CLAIM 3.2. *Let $\mathcal{V}$ be an irreducible subvariety of $\mathbb{P}^n$. There exists a subvariety $\widetilde{\mathcal{V}} \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree at most $\deg(\mathcal{V})$ such that any k-subspace of $\mathbb{P}^n$ that evades $\widetilde{\mathcal{V}}$ also evades $\mathcal{V}$.*

PROOF. If $\dim(\mathcal{V}) = n-k-1$, then just let $\widetilde{\mathcal{V}} = \mathcal{V}$.

Now assume $\dim(\mathcal{V}) < n-k-1$. Let $t = (n-k-1) - \dim(\mathcal{V}) - 1$ and let $\widetilde{\mathcal{V}}$ be the join of $\mathcal{V}$ and a t-subspace disjoint from $\mathcal{V}$ (which exists by Corollary 2.8). Then $\widetilde{\mathcal{V}}$ is a projective subvariety of dimension $n-k-1$ and degree at most $\deg(\mathcal{V})$ by Lemma 2.6. Suppose W is a k-subspace that evades $\widetilde{\mathcal{V}}$. Then W is disjoint from $\widetilde{\mathcal{V}} \supseteq \mathcal{V}$. So W also evades $\mathcal{V}$.

Finally, assume $\dim(\mathcal{V}) > n-k-1$. Let $t = \dim(\mathcal{V}) - (n-k-1)$. By Lemma 2.7, there exist t hyperplanes $H_1, \ldots, H_t$ of $\mathbb{P}^n$ such that $\mathcal{V} \cap \bigcap_{i=1}^t H_i$ is equidimensional of dimension $n-k-1$ and degree at most $\deg(\mathcal{V})$. Let $\widetilde{\mathcal{V}} = \mathcal{V} \cap \bigcap_{i=1}^t H_i$. Suppose W is a k-subspace that evades $\widetilde{\mathcal{V}}$. Then $W \cap \widetilde{\mathcal{V}} = (W \cap \mathcal{V}) \cap \bigcap_{i=1}^t H_i = \emptyset$. Again by Lemma 2.7, we have $\dim(W \cap \mathcal{V}) \leq t - 1 = \dim(\mathcal{V}) + \dim(W) - n$. So W also evades $\mathcal{V}$. □

PROOF (Lemma 3.1). Consider a projective subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of degree at most d. Let $\mathcal{V}_1, \ldots, \mathcal{V}_s$ be the irreducible components of $\mathcal{V}$. For each $i \in [s]$, use Claim 3.2 to choose a projective subvariety $\widetilde{\mathcal{V}}_i \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree at most $\deg(\mathcal{V}_i)$ such that any k-subspace that evades $\widetilde{\mathcal{V}}_i$ also evades $\mathcal{V}_i$. Let $\widetilde{\mathcal{V}} = \bigcup_{i=1}^s \widetilde{\mathcal{V}}_i$.

Then $\widetilde{\mathcal{V}} \in \mathcal{F}$. By construction, any k-subspace that evades $\widetilde{\mathcal{V}}$ also evades $\mathcal{V}$. It follows that an $(\mathcal{F}, \varepsilon)$-evasive k-subspace family is also (n, d, ε)-evasive. □

We further reduce to the case of irreducible varieties at the cost of blowing up the parameter ε by a factor of d. This is useful as we need irreducibility later in Lemma 3.13.

LEMMA 3.3. *Let $n, d \in \mathbb{N}^+$ and $k \in \{0, 1, \ldots, n-1\}$. Let $\mathcal{F}'$ be the family of all irreducible subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. Then an $(\mathcal{F}', \varepsilon)$-evasive k-subspace family is also an $(n, d, d\varepsilon)$-evasive k-subspace family.*

PROOF. Let $\mathcal{F}$ be as in Lemma 3.1. Each $\mathcal{V} \in \mathcal{F}$ has at most d irreducible components, which are all in $\mathcal{F}'$ since their degrees are bounded by d. By definition and the union bound, if a k-subspace family $\mathcal{H}$ is $(\mathcal{F}', \varepsilon)$-evasive, then it is also $(\mathcal{F}, d\varepsilon)$-evasive. Combining this with Lemma 3.1 proves the lemma. □

3.2. Chow Forms. By Lemma 3.1 and Lemma 3.3, we only need to evade equidimensional or irreducible projective subvarieties of codimension $k+1$. The "bad" k-subspaces that intersect such a variety $\mathcal{V}$ form a hypersurface of the Grassmannian defined by a single form called the *Chow form* of $\mathcal{V}$. We now explain the basic theory of Chow forms.

Grassmannians. Let $n \in \mathbb{N}$ and $k \in \{0, 1, \ldots, n-1\}$. The *Grassmannian* $\mathrm{G}(k+1, n+1)$ is the set of all $(k+1)$-dimensional linear subspaces of $\mathbb{A}^{n+1}$. By taking the quotient modulo scalars, it may also be identified with the set of all k-subspaces of $\mathbb{P}^n$, which we denote by $\mathbb{G}(k, n)$.

The Plücker embedding and Plücker coordinates. Consider a linear subspace $W \in \mathrm{G}(k+1, n+1)$. The simplest way of representing W is using a $(k+1) \times (n+1)$ matrix A over $\mathbb{F}$ such that W equals the row space of A. We call such a matrix A a *generating matrix* of W. For convenience, we also say A is a generating matrix of $\mathbb{P}W \in \mathbb{G}(k, n)$.

The entries of A are called the *(primal) Stiefel coordinates* of W. However, note that A is not uniquely determined by W since for any $(k+1)\times(k+1)$ invertible matrix M over $\mathbb{F}$, the matrix MA is also a generating matrix of W.

Another way of representing W is using the vector $(\det A_{[k+1],S})_{S\in\binom{[n+1]}{k+1}}$ of maximal minors of a generating matrix A of W. For a $(k+1)\times(k+1)$ invertible matrix M over $\mathbb{F}$, replacing A by MA corresponds to multiplying all the maximal minors $\det A_{[k+1],S}$ by $\det M\in\mathbb{F}^\times$. To remove ambiguity, we could view $(\det A_{[k+1],S})_{S\in\binom{[n+1]}{k+1}}$ as a point in the projective space $\mathbb{P}^{\binom{n+1}{k+1}-1}$, which is then uniquely determined by W. This leads to the definition of the *Plücker embedding*.

Definition 3.4 (Plücker embedding). *Define* $\phi:\mathrm{G}(k+1,n+1)\to\mathbb{P}^{\binom{n+1}{k+1}-1}$ *by*

$$\phi(W)=(\det A_{[k+1],S})_{S\in\binom{[n+1]}{k+1}}$$

where A is a generating matrix of W.

The Plücker embedding embeds the Grassmannian $\mathrm{G}(k+1,n+1)$ in $\mathbb{P}^{\binom{n+1}{k+1}-1}$ as an irreducible projective subvariety, as stated by the following theorem. See, e.g., Fulton (1997); Harris (1992) for proofs.

Theorem 3.5. *The Plücker embedding ϕ is a well-defined injective map whose image is an irreducible projective subvariety of $\mathbb{P}^{\binom{n+1}{k+1}-1}$.*

The homogeneous coordinates $(\det A_{[k+1],S})_{S\in\binom{[n+1]}{k+1}}$ of $\phi(W)$ are called the *(primal) Plücker coordinates* of W.

Denote by $R:=\mathbb{F}\left[X_S:S\in\binom{[n+1]}{k+1}\right]$ the homogeneous coordinate ring of $\mathbb{P}^{\binom{n+1}{k+1}-1}$. The irreducible projective subvariety $\phi(\mathrm{G}(k+1,n+1))$ is defined by a homogeneous prime ideal of R, which we denoted by I. Then R/I is the homogeneous coordinate ring of $\phi(\mathrm{G}(k+1,n+1))$. The ideal I contains precisely the polynomial relations that the Plücker coordinates need to satisfy. It is also

known that I is generated by certain quadratic forms, known as the *Plücker relations*. See Fulton (1997); Harris (1992) for details.

Dual Plücker coordinates. Alternatively, we could represent a linear subspace $W \in \mathrm{G}(k+1, n+1)$ by an $(n-k) \times (n+1)$ matrix B over $\mathbb{F}$ whose rows specify the linear equations defining W. We call such a matrix B a *parity check matrix* of W. For convenience, we also say B is a parity check matrix of $\mathbb{P}W \in \mathbb{G}(k, n)$.

The entries of B are called the *dual Stiefel coordinates* of W. This gives another embedding $\phi^\vee : \mathrm{G}(k+1, n+1) \to \mathbb{P}^{\binom{n+1}{n-k}-1} = \mathbb{P}^{\binom{n+1}{k+1}-1}$, defined by

$$\phi^\vee(W) = (\det B_{[n-k],S})_{S \in \binom{[n+1]}{n-k}}.$$

The homogeneous coordinates $(\det B_{[n-k],S})_{S \in \binom{[n+1]}{n-k}}$ of $\phi^\vee(W)$ are called the *dual Plücker coordinates* of W. In fact, it is known that dual Plücker coordinates are equivalent to primal Plücker coordinates. Namely, if $W \in \mathrm{G}(k+1, n+1)$ has primal Plücker coordinates $(c_S)_{S \in \binom{[n+1]}{k+1}}$, then it has dual Plücker coordinates $(c'_S)_{S \in \binom{[n+1]}{n-k}}$ with $c'_S = (-1)^{\sum_{i \in S} i - \sum_{i \in [k+1]} i} \cdot c_{[n+1] \setminus S}$ (see, e.g., Joswig & Theobald (2013)).

Chow forms. Recall that we denote by $\mathbb{G}(k, n)$ the set of all k-subspaces of $\mathbb{P}^n$. By identifying $\mathrm{G}(k+1, n+1)$ with $\mathbb{G}(k, n)$ via $W \mapsto \mathbb{P}W$, we regard ϕ and $\phi^\vee$ as maps from $\mathbb{G}(k, n)$ to $\mathbb{P}^{\binom{n+1}{k+1}-1}$.

We also need the notion of *associated hypersurfaces*.

Definition 3.6 (Gelfand *et al.* 1994). *For an irreducible subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of dimension $n-k-1$, define the* associated hypersurfaces *$\mathcal{Z}_\mathcal{V}$ of $\mathcal{V}$ as the set of k-subspaces intersecting $\mathcal{V}$, i.e.,*

$$\mathcal{Z}_\mathcal{V} := \{W \in \mathbb{G}(k, n) : \mathcal{V} \cap W \neq \emptyset\}.$$

The term "associated hypersurface" is justified by the following theorem.

THEOREM 3.7. *Let $\mathcal{V} \subseteq \mathbb{P}^n$ be an irreducible projective subvariety of dimension $n-k-1$ and degree $d \in \mathbb{N}^+$. Then there exists a nonzero homogeneous polynomial $P_{\mathcal{V}} \in R = \mathbb{F}\left[X_S : S \in \binom{[n+1]}{k+1}\right]$ of degree d such that $\phi(\mathcal{Z}_{\mathcal{V}})$ is defined by $P_{\mathcal{V}}$ as a subvariety of $\phi(\mathbb{G}(k,n))$. That is,*

$$\phi(\mathcal{Z}_{\mathcal{V}}) = \phi(\mathbb{G}(k,n)) \cap \mathcal{V}(P_{\mathcal{V}}).$$

Moreover, $\mathcal{R}_{\mathcal{V}} := P_{\mathcal{V}} + I \in (R/I)_d$ is uniquely determined by $\mathcal{V}$ up to scalars.

Theorem 3.7 is explicitly stated as Dalbec & Sturmfels (1995, Theorem 1.1 and Corollary 2.1). A proof can be found in Gelfand *et al.* (1994, Section 3.2). We briefly explain how to find a polynomial $P_{\mathcal{V}}$ satisfying Theorem 3.7: Firstly, it can be shown using the trick of dimension counting via incidence varieties that $\phi(\mathcal{Z}_{\mathcal{V}})$ is an irreducible projective subvariety of the Grassmannian $\phi(\mathbb{G}(k,n))$ of codimension one (Gelfand *et al.* 1994, Section 3.2, Proposition 2.2). Secondly, the homogeneous coordinate ring R/I of the Grassmannian is known to be a *unique factorization domain* (Fulton 1997, Chapter 9). These two facts imply that the homogeneous ideal of R/I defining $\phi(\mathcal{Z}_{\mathcal{V}})$ is a *principal ideal*. Choose $\mathcal{R}_{\mathcal{V}}$ to be a generator of this principal ideal, which is unique up to scalars. Then lift $\mathcal{R}_{\mathcal{V}} \in R/I$ to $P_{\mathcal{V}} \in R$.

Now we are ready to define the Chow form of projective subvarieties.

DEFINITION 3.8 (Chow form). *Let $\mathcal{V} \subseteq \mathbb{P}^n$ be an irreducible subvariety of dimension $n-k-1$ and degree $d \in \mathbb{N}^+$. Define the* Chow form of $\mathcal{V}$ in Plücker coordinates, *or simply the* Chow form *of $\mathcal{V}$, to be $\mathcal{R}_{\mathcal{V}} \in (R/I)_d$ as in Theorem 3.7.*

More generally, for an equidimensional subvariety $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i \subseteq \mathbb{P}^n$ of dimension $n-k-1$ and degree d, where $\mathcal{V}_1, \ldots, \mathcal{V}_s$ are the irreducible components of $\mathcal{V}$, the Chow form *of $\mathcal{V}$ is $\mathcal{R}_{\mathcal{V}} := \prod_{i=1}^s \mathcal{R}_{\mathcal{V}_i} \in (R/I)_d$. It is uniquely determined by $\mathcal{V}$ up to scalars.*

As a k-subspace intersects $\mathcal{V} = \bigcup_{i=1}^s \mathcal{V}_i$ iff it intersects some $\mathcal{V}_i$, we see from Theorem 3.7 that the Chow form $\mathcal{R}_{\mathcal{V}}$ of an equidimensional projective subvariety $\mathcal{V}$ of dimension $n-k-1$ vanishes precisely at the set of k-subspaces that intersect $\mathcal{V}$.

EXAMPLE 3.9. Let $k = 0$. Let $\mathcal{V} \subseteq \mathbb{P}^n$ be a hypersurface defined by a nonzero homogeneous polynomial $P \in \mathbb{F}[X_1, \ldots, X_{n+1}] = R$. The ideal I of R is zero in this case. And the Chow form $\mathcal{R}_\mathcal{V}$ of $\mathcal{V}$ is simply P (up to a scalar). ◊

EXAMPLE 3.10. Let $V \in \mathrm{G}(n-k, n+1)$ and $W \in \mathrm{G}(k+1, n+1)$. Choose matrices $A, B \in \mathbb{F}^{(k+1)\times(n+1)}$ such that A is a generating matrix of W and B is a parity check matrix of V. Then $\mathbb{P}V \cap \mathbb{P}W \neq \emptyset$ iff $\dim(V \cap W) > 0$, which holds iff $\det(AB^T) = 0$. On the other hand, we have

$$\begin{aligned}\det(AB^T) &= \sum_{S \in \binom{[n+1]}{k+1}} \det(A_{[k+1],S}) \cdot \det((B^T)_{S,[k+1]}) \\ &= \sum_{S \in \binom{[n+1]}{k+1}} \det(A_{[k+1],S}) \cdot \det(B_{[k+1],S}),\end{aligned}$$

where the first equation is known as the *Cauchy–Binet formula* (see, e.g., Forbes *et al.* (2014)). So $P_{\mathbb{P}V} \in R_1$ is a linear polynomial whose coefficients are given by the dual Plücker coordinates $(\det B_{[k+1],S})_{S \in \binom{[n+1]}{k+1}}$ of V (up to a scalar). The degree-one part I_1 of I is zero as I is generated by quadratic forms. So the Chow form $\mathcal{R}_{\mathbb{P}V} \in (R/I)_1 = R_1$ is simply $P_{\mathbb{P}V}$. ◊

Chow forms in Stiefel coordinates. We may also express the Chow form in Stiefel coordinates, i.e., in the entries of a generating matrix of a linear subspace. This expression has the advantage that it is an actual polynomial rather than a member of the abstract vector space $(R/I)_d$.

Formally, let A^* be a $(k+1) \times (n+1)$ variable matrix whose (i,j)-th entry is a variable $Y_{i,j}$. Define the ring homomorphism

$$\phi^\sharp : R = \mathbb{F}\left[X_S : S \in \binom{[n+1]}{k+1}\right] \to \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$$

that sends each variable X_S to $\det(A^*_{[k+1],S})$. Define the *Chow form of $\mathcal{V}$ in Stiefel coordinates* to be

$$\widetilde{\mathcal{R}}_\mathcal{V} := \phi^\sharp(P_\mathcal{V}) \in \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$$

where $P_{\mathcal{V}} \in R_d$ is a lift of $\mathcal{R}_{\mathcal{V}} \in (R/I)_d$. Note that I is precisely the kernel of $\phi^\sharp$. So $\widetilde{\mathcal{R}}_{\mathcal{V}}$ is uniquely determined by $\mathcal{V}$ up to scalars. By construction, for any $W \in \mathrm{G}(k+1, n+1)$ and generating matrix $A = (a_{i,j})_{i\in[k+1], j\in[n+1]}$ of W, we have $P_{\mathcal{V}}(\phi(W)) = \widetilde{\mathcal{R}}_{\mathcal{V}}(A) := \widetilde{\mathcal{R}}_{\mathcal{V}}(a_{1,1}, \ldots, a_{k+1,n+1})$. So $\widetilde{\mathcal{R}}_{\mathcal{V}}$ vanishes at A iff $\mathbb{P}W \in \mathbb{G}(k, n)$ intersects $\mathcal{V}$.

Chow forms in dual Stiefel coordinates. Similarly, we may express the Chow form in dual Stiefel coordinates, i.e., in the entries of a parity check matrix of a linear subspace.

More specifically, choose a homogeneous polynomial $Q_{\mathcal{V}} \in \mathbb{F}\left[X_S : S \in \binom{[n+1]}{n-k}\right]$ that defines the set of k-subspaces intersecting $\mathcal{V}$ in terms of dual Plücker coordinates. As primal and dual Plücker coordinates are equivalent, $Q_{\mathcal{V}}$ can be obtained from the polynomial $P_{\mathcal{V}}$ above by simply negating and renaming variables. Next, compose $Q_{\mathcal{V}}$ with a ring homomorphism that substitutes dual Plücker coordinates with dual Stiefel coordinates. The resulting polynomial, which we denote by $\widetilde{\mathcal{R}}^{\vee}_{\mathcal{V}} \in \mathbb{F}[Y_{i,j} : i \in [n-k], j \in [n+1]]$, is called the *Chow form of* $\mathcal{V}$ *in dual Stiefel coordinates*.

We note that the Chow form $\widetilde{\mathcal{R}}_{\mathcal{V}}$ in primal Stiefel coordinates is a homogeneous polynomial of degree $(k+1)d$ in $(k+1)(n+1)$ variables, whereas the Chow form $\widetilde{\mathcal{R}}^{\vee}_{\mathcal{V}}$ in dual Stiefel coordinates is a homogeneous polynomial of degree $(n-k)d$ in $(n-k)(n+1)$ variables. This suggests that it is more convenient to use the Chow form in primal (resp. dual) Stiefel coordinates when k is small (resp. $n-k$ is small).

3.3. Explicit Constructions of Variety Evasive Subspace Families. Let $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\varepsilon \in (0, 1)$. In this subsection, we prove the Main theorem (Theorem 1.7) by constructing explicit projective or affine k-subspace families that are (n, d, ε)-evasive.

We first prove Theorem 1.7 in the projective case and then derive the affine case from it by viewing $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$. For the projective case, we present two constructions. The first one is simple and only uses ε-hitting sets for low degree polynomials (Lemma 2.3). But the size of the resulting subspace family is

polynomial only when both d and k (or $n-k$) are bounded. Next, we give a more sophisticated construction, which yields subspace families of polynomial size as long as d is bounded.

3.3.1. Simple Construction. We first present a simple construction of (n, d, ε)-evasive k-subspace families on $\mathbb{P}^n$.

First assume $k+1 \leq n-k$. In this case, construct a k-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ as follows:

1. Use Lemma 2.3 to compute an ε-hitting set T for the family of polynomials $f \in \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$ of degree at most $(k+1)d$ such that $|T| = \text{poly}\left(\binom{(k+1)(n+1+d)}{(k+1)d}, 1/\varepsilon\right)$. Think of T as a collection of $(k+1) \times (n+1)$ matrices over $\mathbb{F}$.

2. Initialize $\mathcal{H} = \emptyset$. For each matrix $A \in T$, if A has full row rank $k+1$, add to $\mathcal{H}$ the k-subspace $W \in \mathbb{G}(k, n)$ with the generating matrix A.

Next, assume $k+1 > n-k$. In this case, construct $\mathcal{H}$ in a similar way, but use parity check matrices instead of generating matrices. Namely, compute an ε-hitting set T for the family of polynomials $f \in \mathbb{F}[Y_{i,j} : i \in [n-k], j \in [n+1]]$ of degree at most $(n-k)d$ such that $|T| = \text{poly}\left(\binom{(n-k)(n+1+d)}{(n-k)d}, 1/\varepsilon\right)$. Think of T as a collection of $(n-k) \times (n+1)$ matrices over $\mathbb{F}$. For each matrix $A \in T$, add to $\mathcal{H}$ the k-subspace $W \in \mathbb{G}(k, n)$ with the parity check matrix A.

This construction does give an (n, d, ε)-evasive k-subspace family, as stated by the following lemma.

LEMMA 3.11. *The k-subspace family $\mathcal{H}$ constructed above is (n, d, ε)-evasive and has size polynomial in*

$$\min\left\{\binom{(k+1)(n+1+d)}{(k+1)d}, \binom{(n-k)(n+1+d)}{(n-k)d}\right\}$$

and $1/\varepsilon$. Moreover, the total time complexity of computing the linear equations defining the k-subspaces in $\mathcal{H}$ is polynomial in $|\mathcal{H}|$ (and $\log p$, if $\text{char}(\mathbb{F}) = p > 0$).

Proof. We only show that $\mathcal{H}$ is (n, d, ε)-evasive since the rest of the lemma is obvious from the construction. Let $\mathcal{F}$ be the family of all equidimensional subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. By Lemma 3.1, it suffices to prove that $\mathcal{H}$ is $(\mathcal{F}, \varepsilon)$-evasive. Consider any $\mathcal{V} \in \mathcal{F}$. We want to show that $\mathcal{V} \cap W = \emptyset$ for all but at most ε-fraction of $W \in \mathcal{H}$.

First assume $k+1 \le n-k$, or equivalently, $\binom{(k+1)(n+1+d)}{(k+1)d} \le \binom{(n-k)(n+1+d)}{(n-k)d}$. The Chow form $\widetilde{\mathcal{R}}_{\mathcal{V}}$ of $\mathcal{V}$ in Stiefel coordinates is a nonzero homogeneous polynomial in $\mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$ of degree $(k+1)\deg(\mathcal{V}) \le (k+1)d$. By the choice of T, for all but at most ε-fraction of $A \in T$, we have $\widetilde{\mathcal{R}}_{\mathcal{V}}(A) \neq 0$, which implies $\mathcal{V} \cap W = \emptyset$, where A is a generating matrix of W.

By construction, $\mathcal{H}$ is the collection of k-subspaces corresponding to the matrices $A \in T$ of full row rank. So we have ignored the matrices that do not have full row rank. But this does not increase the fraction of "bad" $W \in \mathcal{H}$ since if A does not have full row rank, then the maximal minors of A are all zero, and $\widetilde{\mathcal{R}}_{\mathcal{V}}(A)$ must be zero. It follows that $\mathcal{V} \cap W = \emptyset$ for all but at most ε-fraction of $W \in \mathcal{H}$, as desired.

Now assume $k+1 > n-k$. The proof in this case is similar and we omit the details. The only difference is that we use the Chow form $\widetilde{\mathcal{R}}^{\vee}_{\mathcal{V}}$ in dual Stiefel coordinates instead of $\widetilde{\mathcal{R}}_{\mathcal{V}}$. □

3.3.2. Improved Construction. Before presenting the improved construction, we first introduce some notions from algebraic geometry.

Projections. Suppose W is a k-subspace of $\mathbb{P}^n$, and $\ell_1, \ldots, \ell_{n-k} \in \mathbb{F}[X_1, \ldots, X_{n+1}]$ are $n-k$ homogeneous linear polynomials such that $W = \mathcal{V}(\ell_1, \ldots, \ell_{n-k})$. Then we have a map $\pi : \mathbb{P}^n \setminus W \to \mathbb{P}^{n-k-1}$ defined by

$$\pi : \mathbf{x} \mapsto (\ell_1(\mathbf{x}), \ldots, \ell_{n-k}(\mathbf{x}))$$

which is well-defined since $\ell_1, \ldots, \ell_{n-k}$ never simultaneously vanish on $\mathbb{P}^n \setminus W$. We say π is a *projection* from $\mathbb{P}^n \setminus W$ to $\mathbb{P}^{n-k-1}$ and W is its *center*. Note that if we lift π to the linear map $\pi' : \mathbb{A}^{n+1} \to \mathbb{A}^{n-k}$

sending $\mathbf{x} \in \mathbb{A}^{n+1}$ to $(\ell_1(\mathbf{x}), \ldots, \ell_{n-k}(\mathbf{x})) \in \mathbb{A}^{n-k}$, then the center W is simply $\mathbb{P}\ker(\pi')$.

We need the following lemma.

LEMMA 3.12 (Shafarevich 1994, Section I.5.3, Theorem 7). *Suppose $\pi : \mathbb{P}^n \setminus W \to \mathbb{P}^m$ is a projection with center W and $\mathcal{V}$ is a subvariety of $\mathbb{P}^n$ disjoint from W. Then π restricts to a finite morphism from $\mathcal{V}$ to $\mathbb{P}^m$.*

Nondegenerate varieties. For a subvariety $\mathcal{V} \subseteq \mathbb{P}^n$, denote by $\mathrm{span}(\mathcal{V})$ the smallest projective subspace that contains $\mathcal{V}$. We say $\mathcal{V}$ is *nondegenerate* if it is not contained in a hyperplane of $\mathbb{P}^n$, or equivalently, $\mathrm{span}(\mathcal{V}) = \mathbb{P}^n$.

We need the following fact from algebraic geometry (see, e.g., Eisenbud & Harris (1987, Proposition 0) or Harris (1992, Corollary 18.12)).

LEMMA 3.13. *The codimension of a nondegenerate irreducible subvariety $\mathcal{V}$ of $\mathbb{P}^n$ is at most* $\deg(\mathcal{V}) - 1$.

A two-step construction. We now give an improved construction of (n, d, ε)-evasive k-subspace families on $\mathbb{P}^n$ as follows.

1. If $k \leq d - 2$, just use the previous simple construction. So assume $k > d - 2$. Let $k' = d - 2 < k$, $n' = k' + n - k < n$, and $\varepsilon_0 = \varepsilon/(2d)$.

2. Use Corollary 2.5 to construct a collection $\mathcal{H}_1$ of $(n'+1) \times (n+1)$ matrix over $\mathbb{F}$ such that $|\mathcal{H}_1| = \mathrm{poly}(n, d/\varepsilon)$ and for every $(n+1) \times (n'+1)$ matrix M over $\mathbb{F}$ of rank $n'+1$, all but at most ε_0-fraction of $B \in U$ satisfies $\mathrm{rank}(BM) = n' + 1$.

   We abuse the notation and view $\mathcal{H}_1$ as a collection of linear maps from $\mathbb{A}^{n+1}$ to $\mathbb{A}^{n'+1}$. Then for any linear subspace $W \subseteq \mathbb{A}^{n+1}$ of dimension $n' + 1$, we have $\dim(\pi(W)) = \dim(W) = n' + 1$ for all but at most ε_0-fraction of $\pi \in \mathcal{H}_1$.

3. Construct a collection $\mathcal{H}_2$ of linear maps from $\mathbb{A}^{n'+1}$ to $\mathbb{A}^{n-k}$ as follows. First assume $d > 1$. Use Lemma 3.11 to construct

an (n', d, ε_0)-evasive k'-subspace family $\mathcal{H}_2'$ on $\mathbb{P}^{n'}$ of size polynomial in $\min\left\{\binom{(k'+1)(n'+1+d)}{(k'+1)d}, \binom{(n-k)(n'+1+d)}{(n-k)d}\right\}$ and $1/\varepsilon_0$. For each k'-subspace $W \in \mathcal{H}_2$, compute a surjective linear map $\pi_W : \mathbb{A}^{n'+1} = \mathbb{A}^{n-k+k'+1} \to \mathbb{A}^{n-k}$ such that $W = \mathbb{P}\ker(\pi_W)$. Let $\mathcal{H}_2 = \{\pi_W : W \in \mathcal{H}_2'\}$.

If $d = 1$, just let $\mathcal{H}_2$ be the singleton consisting of the identity map on $\mathbb{A}^{n'+1} = \mathbb{A}^{n-k}$.

4. Initialize $\mathcal{H} = \emptyset$. For each $(\pi_1, \pi_2) \in \mathcal{H}_1 \times \mathcal{H}_2$, if $\dim(\ker(\pi_2 \circ \pi_1)) = k + 1$, add the k-subspace $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ to $\mathcal{H}$.

We use the construction above to prove the Main theorem (Theorem 1.7) in the projective case. For convenience, we restate it in the following form.

THEOREM 3.14 (Main theorem in the projective case). *The k-subspace family $\mathcal{H}$ constructed above is (n, d, ε)-evasive and has size* $\mathrm{poly}(N(k, d, n), n, 1/\varepsilon)$. *Moreover, the total time complexity of computing the linear equations defining the k-subspaces in $\mathcal{H}$ is polynomial in $|\mathcal{H}|$ (and* $\log p$, *if* $\mathrm{char}(\mathbb{F}) = p > 0$*).*

PROOF. The theorem follows from Lemma 3.11 if $k \le d - 2$. So assume $k > d - 2$. We only show that $\mathcal{H}$ is (n, d, ε)-evasive since the rest of the theorem is obvious from the construction.

Let $\mathcal{F}$ be the family of all irreducible subvarieties of $\mathbb{P}^n$ of dimension $n - k - 1$ and degree at most d. By Lemma 3.3, it suffices to prove that $\mathcal{H}$ is $(\mathcal{F}, 2\varepsilon_0)$-evasive. Consider any $\mathcal{V} \in \mathcal{F}$. We want to show that $\mathcal{V} \cap W = \emptyset$ for all but at most $(2\varepsilon_0)$-fraction of $W \in \mathcal{H}$.

By definition, $\mathcal{V}$ is a nondegenerate irreducible subvariety of $\mathrm{span}(\mathcal{V})$. By Lemma 3.13, the codimension of $\mathcal{V}$ in $\mathrm{span}(\mathcal{V})$ is at most $d - 1$. Therefore,

$$\dim(\mathrm{span}(\mathcal{V})) \le \dim(\mathcal{V}) + d - 1 = (n - k - 1) + (d - 1) = n'.$$

Let $\Lambda \subseteq \mathbb{P}^n$ be an n'-subspace that contains $\mathrm{span}(\mathcal{V})$. By the choice of $\mathcal{H}_1$, all but at most ε_0-fraction of $\pi_1 \in \mathcal{H}_1$ satisfies $\mathbb{P}\ker(\pi_1) \cap \Lambda = \emptyset$. Fix π_1 that satisfies this condition. Then $\mathbb{P}\ker(\pi_1)$ is disjoint

from $\mathcal{V} \subseteq \Lambda$. By Lemma 3.12, π_1 induces a finite morphism $\bar{\pi}_1 : \mathcal{V} \to \mathbb{P}^{n'}$.

Let $\mathcal{V}' = \bar{\pi}_1(\mathcal{V}) \subseteq \mathbb{P}^{n'}$. Then $\mathcal{V}'$ is a projective subvariety of dimension $\dim(\mathcal{V}) = n - k - 1$ and degree at most d. (The degree bound follows from, e.g., an inductive application of Mumford (1976, Proposition 5.5).) By the choice of $\mathcal{H}_2$, all but at most ε_0-fraction of $\pi_2 \in \mathcal{H}_2$ satisfies $\mathbb{P}\ker(\pi_2) \cap \mathcal{V}' = \emptyset$. Fix π_2 that satisfies this condition. (If $d = 1$ and π_2 is the identity map, we regard $\mathbb{P}\ker(\pi_2)$ as an empty set, in which case this condition is also satisfied.) By Lemma 3.12, π_2 induces a finite morphism $\bar{\pi}_2 : \mathcal{V}' \to \mathbb{P}^{n-k-1}$. So we have a finite morphism $\bar{\pi}_2 \circ \bar{\pi}_1 : \mathcal{V} \to \mathbb{P}^{n-k-1}$. Note that $\bar{\pi}_2 \circ \bar{\pi}_1$ is defined by restricting a projection with center $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ to $\mathcal{V}$. As $\bar{\pi}_2 \circ \bar{\pi}_1$ is well-defined on $\mathcal{V}$, its center $\mathbb{P}\ker(\pi_2 \circ \pi_1)$ is disjoint from $\mathcal{V}$ and this also forces $\dim \ker(\pi_2 \circ \pi_1) = k + 1$ by Lemma 2.9.

By the above argument and the construction of $\mathcal{H}$, all but at most $2\varepsilon_0$-fraction of the k-subspaces in $\mathcal{H}$ are disjoint from $\mathcal{V}$, as desired. □

3.3.3. Derandomizing Noether's Normalization Lemma.

We now prove Theorem 1.12 and Theorem 1.13. For convenience, we restate the theorems below.

THEOREM 1.12. *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\varepsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^{n+1} \to \mathbb{A}^{r+1}$ of size $\mathrm{poly}(N(k, d, n), n, 1/\varepsilon)$ such that for every subvariety $\mathcal{V} \subseteq \mathbb{P}^n$ of dimension r and degree at most d, all but at most ε-fraction of $\pi \in \mathcal{L}$ induce a surjective finite morphism from $\mathcal{V}$ to $\mathbb{P}^r$.*[3] *Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

PROOF. If $r = n$, just use the identity map on $\mathbb{A}^{n+1} = \mathbb{A}^{r+1}$. So assume $r < n$. Use Theorem 3.14 to construct an (n, d, ε)-evasive k-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ of size $\mathrm{poly}(N(k, d, n), n, 1/\varepsilon)$. For each k-subspace $W \in \mathcal{H}$, compute a surjective linear map $\pi_W : \mathbb{A}^{n+1} \to$

[3] Let $N(k, d, n) = 1$ when $r = n$ (i.e., $k = -1$). Similarly, in Theorem 1.13, let $N(k, d, n - 1) = 1$ when $r = n$ or $r = 0$ (i.e., $k = -1$ or $k = n - 1$).

$\mathbb{A}^{r+1}$ such that $W = \mathbb{P}\ker(\pi_W)$. Let $\mathcal{L} = \{\pi_W : W \in \mathcal{H}\}$. Then $\mathcal{L}$ is a desired collection of linear maps by Lemma 3.12. □

THEOREM 1.13. *Let $n, d \in \mathbb{N}^+$, $r \in \{0, 1, \ldots, n\}$, $k = n - r - 1$, and $\varepsilon \in (0, 1)$. There exists an explicit collection $\mathcal{L}$ of linear maps $\mathbb{A}^n \to \mathbb{A}^r$ of size $\mathrm{poly}(N(k, d, n-1), n, 1/\varepsilon)$ such that for every subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of dimension r and degree at most d, all but at most ε-fraction of $\pi \in \mathcal{L}$ restrict to a surjective finite morphism from $\mathcal{V}$ to $\mathbb{A}^r$. Moreover, $\mathcal{L}$ can be computed in time polynomial in $|\mathcal{L}|$ (and $\log p$, if $\mathrm{char}(\mathbb{F}) = p > 0$).*

PROOF. If $r = n$, just use the identity map on $\mathbb{A}^n = \mathbb{A}^r$. If $r = 0$, use the only map $\mathbb{A}^n \to \mathbb{A}^0$. So assume $0 < r < n$. Regard $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$ via $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. Similarly, regard $\mathbb{A}^r$ as an open subset of $\mathbb{P}^r$ via $(x_1, \ldots, x_r) \mapsto (x_1, \ldots, x_r, 1)$. Let H_∞ be the hyperplane at infinity of $\mathbb{P}^n$ defined by $X_{n+1} = 0$.

Use Theorem 3.14 to construct an $(n-1, d, \varepsilon)$-evasive k-subspace family $\mathcal{H}$ on $H_\infty \cong \mathbb{P}^{n-1}$ of size $\mathrm{poly}(N(k, d, n-1), n, 1/\varepsilon)$. For each $W \in \mathcal{H}$, choose $n - k = r + 1$ homogeneous linear polynomials $\ell_1, \ldots, \ell_{r+1} \in \mathbb{F}[X_1, \ldots, X_{n+1}]$ such that $\ell_{r+1} = X_{n+1}$, $\ell_1, \ldots, \ell_r \in \mathbb{F}[X_1, \ldots, X_n]$, and $W = \mathcal{V}(\ell_1, \ldots, \ell_{r+1})$. This is possible as $W \subseteq H_\infty = \mathcal{V}(X_{n+1})$. These $r + 1$ linear polynomials determine a projection $\pi_W : \mathbb{P}^n \setminus W \to \mathbb{P}^r$, defined by

$$\mathbf{x} = (x_1, \ldots, x_{n+1}) \mapsto (\ell_1(\mathbf{x}), \ldots, \ell_{r+1}(\mathbf{x})) = (\ell_1(\mathbf{x}), \ldots, \ell_r(\mathbf{x}), x_{n+1}).$$

As $x_{n+1} = 1$ for $\mathbf{x} \in \mathbb{A}^n$, we have $\pi_W(\mathbb{A}^n) \subseteq \mathbb{A}^r$. Restricting π_W on $\mathbb{A}^n$ yields a map $\pi_W|_{\mathbb{A}^n} : \mathbb{A}^n \to \mathbb{A}^r$, which is a linear map as $\ell_1, \ldots, \ell_r$ are homogeneous linear polynomials in $\mathbb{F}[X_1, \ldots, X_n]$. Let $\mathcal{L} = \{\pi_W|_{\mathbb{A}^n} : W \in \mathcal{H}\}$.

Let $\mathcal{V}$ be a subvariety of $\mathbb{A}^n$ of dimension r and degree at most d. Its projective closure $\mathcal{V}_{\mathrm{cl}}$ has dimension $\dim(\mathcal{V}) = r$ and degree $\deg(\mathcal{V}) \le d$. By the definition of $\mathcal{V}_{\mathrm{cl}}$, none of the irreducible components of $\mathcal{V}_{\mathrm{cl}}$ is fully contained in H_∞. So by Lemma 2.7, the projective subvariety $\mathcal{V}_{\mathrm{cl}} \cap H_\infty$ has dimension $r - 1$ and degree at most d.

By the choice of $\mathcal{H}$, all but at most ε-fraction of $W \in \mathcal{H}$ are disjoint from $\mathcal{V}_{\mathrm{cl}} \cap H_\infty$ and hence from $\mathcal{V}_{\mathrm{cl}}$. So we just need to

prove that for every $W \in \mathcal{H}$ disjoint from $\mathcal{V}_{\text{cl}}$ and the corresponding projection π_W, the map $\pi_W|_{\mathcal{V}} : \mathcal{V} \to \mathbb{A}^r$ is a surjective finite morphism. This follows from Lemma 3.12 and the fact that $\mathcal{V} = \mathcal{V}_{\text{cl}} \cap (\pi_W)^{-1}(\mathbb{A}^r)$. □

3.3.4. Proof of the Main theorem in the Affine Case. We now prove Theorem 1.7 in the affine case. Recall that we may view $\mathbb{A}^n$ as an open subset of $\mathbb{P}^n$ via the map $(x_1, \ldots, x_n) \mapsto (x_1, \ldots, x_n, 1)$. In this way, $\mathbb{P}^n$ becomes the disjoint union of $\mathbb{A}^n$ and the hyperplane at infinity H_∞ defined by $X_{n+1} = 0$.

We use the following lemma to reduce the affine case to the projective case.

LEMMA 3.15. *Let $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, and $\varepsilon' \in (0, 1/2)$. Suppose $\mathcal{H}$ is an (n, d, ε')-evasive k-subspace family on $\mathbb{P}^n$. Then*

$$\mathcal{H}' = \{W \cap \mathbb{A}^n : W \in \mathcal{H}, W \not\subseteq H_\infty\}$$

is an (n, d, ε)-evasive affine k-subspace family on $\mathbb{A}^n$, where $\varepsilon = \varepsilon'/(1-\varepsilon') \leq 2\varepsilon'$. Moreover,

$$\mathcal{H}'' = \{W \in \mathcal{H} : W \not\subseteq H_\infty\} = \{W_{\text{cl}} : W \in \mathcal{H}'\}$$

is an (n, d, ε)-evasive k-subspace family on $\mathbb{P}^n$.

PROOF. By (n, d, ε')-evasiveness of $\mathcal{H}$, at most ε'-fraction of $W \in \mathcal{H}$ are fully contained in H_∞. Throwing away those k-subspaces fully contained in H_∞ increases the error parameter ε' by at most a factor of $1/(1-\varepsilon')$. Therefore, $\mathcal{H}'' = \{W \in \mathcal{H} : W \not\subseteq H_\infty\}$ is (n, d, ε)-evasive. We want to prove that $\mathcal{H}' = \{W \cap \mathbb{A}^n : W \in \mathcal{H}''\}$ is also (n, d, ε)-evasive.

Consider a subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of degree at most d. Let $\mathcal{V}_1, \ldots, \mathcal{V}_s$ be the irreducible components of $\mathcal{V}$. The projective closure $\mathcal{V}_{\text{cl}}$ of $\mathcal{V}$ has the irreducible components $(\mathcal{V}_1)_{\text{cl}}, \ldots, (\mathcal{V}_s)_{\text{cl}}$. Consider a k-subspace $W \in \mathcal{H}''$ that evades $\mathcal{V}_{\text{cl}}$. We just need to prove that $W \cap \mathbb{A}^n$ evades $\mathcal{V}$. This is true since for each $i \in [s]$,

$$\begin{aligned}
&\dim((W \cap \mathbb{A}^n) \cap \mathcal{V}_i) \leq \dim(W \cap (\mathcal{V}_i)_{\text{cl}}) \\
&\leq \dim(W) + \dim((\mathcal{V}_i)_{\text{cl}}) - n = \dim(W \cap \mathbb{A}^n) + \dim(\mathcal{V}_i) - n
\end{aligned}$$

where the second inequality holds since W evades $\mathcal{V}_{\text{cl}}$ and the last equality uses the fact $W \not\subseteq H_\infty$. □

The affine case of Theorem 1.7 now follows easily.

Proof (Theorem 1.7 in the affine case). If $k = n$, just choose $\mathcal{H} = \mathbb{A}^n$. Now assume $k < n$. Construct an $(n, d, \varepsilon/2)$-evasive k-subspace family $\mathcal{H}$ on $\mathbb{P}^n$ using Theorem 3.14. Then

$$\mathcal{H}' := \{W \cap \mathbb{A}^n : W \in \mathcal{H}, W \not\subseteq H_\infty\}$$

is an (n, d, ε)-evasive affine k-subspace family on $\mathbb{A}^n$ by Lemma 3.15. The nonhomogeneous linear equations defining $W \cap \mathbb{A}^n \in \mathcal{H}'$ can be easily computed from the homogeneous linear equations defining $W \in \mathcal{H}$ by letting $X_{n+1} = 1$. □

The proof of Theorem 1.7 is now complete.

Strengthening Theorem 1.7 in the affine case. For projective subvarieties $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathbb{P}^n$ such that $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) \geq n$, the minimum possible dimension of $\mathcal{V}_1 \cap \mathcal{V}_2$ is $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n$, as stated by Lemma 2.9. Nevertheless, for two affine subvarieties $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathbb{A}^n$, it is possible that the intersection of $\mathcal{V}_1$ and $\mathcal{V}_2$ is empty even if its expected dimension $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n$ is nonnegative. For example, the intersection of two distinct and parallel affine hyperplanes $\mathcal{V}_1, \mathcal{V}_2 \subseteq \mathbb{A}^n$ is always empty even if $n \geq 2$. The reason this happens is that, while the dimension of $(\mathcal{V}_1)_{\text{cl}} \cap (\mathcal{V}_2)_{\text{cl}}$ is $n - 2$ (as expected), this intersection is fully contained in the hyperplane H_∞, which is excluded from $\mathbb{A}^n$.

One may strengthen the definition of evading (Definition 1.1) by requiring the intersection of $\mathcal{V}_1$ with every irreducible component of $\mathcal{V}_2$ to have *exactly* the expected dimension. It is possible to construct explicit affine k-subspace families satisfying Theorem 1.7 even under this stronger definition of evading. We sketch the ideas as follows but omit the details.

First construct an $(n-1, d, \varepsilon')$-evasive $(k-1)$-subspace family $\mathcal{H}'$ on $H_\infty \cong \mathbb{P}^{n-1}$ for some sufficiently small ε' depending on ε. Then extend each $W \in \mathcal{H}'$ to a collection of k-subspaces by picking $p \in \mathbb{A}^n$ and taking the k-subspace $J(W, p)$, where the coordinates

of p are chosen from an ε'-hitting set for polynomials of degree at most d given by Lemma 3.11. Call the resulting k-subspace family $\mathcal{H}$. It is easy to prove that $\mathcal{H}$ is $(n, d, O(\varepsilon'))$-evasive.

Furthermore, the affine k-subspace family $\{W \cap \mathbb{A}^n : W \in \mathcal{H}\}$ is (n, d, ε)-evasive even under the stronger definition of evading. To see this, consider an affine subvariety $\mathcal{V} \subseteq \mathbb{A}^n$ of degree at most d. For most $W \in \mathcal{H}$, we have:

- For each irreducible component $\mathcal{V}_i$ of $\mathcal{V}$, the dimension of $(\mathcal{V}_i)_{\mathrm{cl}} \cap W$ is as expected by $(n, d, O(\varepsilon'))$-evasiveness of $\mathcal{H}$ and Lemma 2.9. Call this dimension d_i, which is $-\infty$ if $(\mathcal{V}_i)_{\mathrm{cl}} \cap W = \emptyset$.
- Moreover, the dimension of $((\mathcal{V}_i)_{\mathrm{cl}} \cap H_\infty) \cap (W \cap H_\infty)$ is at most $d_i - 1$ by $(n-1, d, \varepsilon')$-evasiveness of $\mathcal{H}'$.
- Therefore, $\mathcal{V}_i \cap (W \cap \mathbb{A}^n)$ has the expected dimension d_i for each irreducible component $\mathcal{V}_i$ of $\mathcal{V}$.

4. Lower Bound

We prove Theorem 1.10 in this section. The main tool is the notion of *Chow varieties*, which parameterize projective subvarieties. More precisely, they parametrize a generalization of projective subvarieties, called *(effective) algebraic cycles* on a projective space.

Algebraic cycles. An *algebraic r-cycle* (or simply *r-cycle*) on $\mathbb{P}^n$ is a formal linear combination $D = \sum c_i \mathcal{V}_i$ of finitely many irreducible subvarieties $\mathcal{V}_i \subseteq \mathbb{P}^n$ of dimension r, where the coefficients c_i are integers. The *degree* of D is $\deg(D) := \sum c_i \deg(\mathcal{V}_i)$. The *support* of D is $\mathrm{supp}(D) := \bigcup_{c_i \neq 0} \mathcal{V}_i$. An r-cycle is *effective* if all its coefficients are nonnegative. Denote by $C(r, d, n)$ the set of all effective r-cycles of degree d on $\mathbb{P}^n$.

Chow varieties. Let $k \in \{0, 1, \ldots, n-1\}$ and $r = n-k-1$. The definition of Chow forms naturally extends to effective r-cycles. Namely, for an effective r-cycle $D = \sum_{i=1}^{r} c_i \mathcal{V}_i$ of degree d on $\mathbb{P}^n$, define the Chow form of D to be $\mathcal{R}_D := \prod_{i=1}^{r} \mathcal{R}_{\mathcal{V}_i}^{c_i}$.

Note that $\mathcal{R}_D$ is a vector in $(R/I)_d$ and is uniquely determined by D up to scalars. Write $[\mathcal{R}_D]$ for the point in $\mathbb{P}(R/I)_d$ represented by $\mathcal{R}_D$. Then we have map $\psi : C(r,d,n) \to \mathbb{P}(R/I)_d$, given by

$$\psi : D \mapsto [\mathcal{R}_D],$$

called the *Chow embedding* of $C(r,d,n)$. Indeed, it embeds $C(r,d,n)$ in $\mathbb{P}(R/I)_d$ as a projective subvariety, as stated by the following theorem.

THEOREM 4.1 (Chow & van der Waerden 1937). *The map ψ is injective and its image is Zariski-closed.*

A proof can also be found in Gelfand *et al.* (1994, Chapter 4). We identify $C(r,d,n)$ with its image under ψ and view it as a projective variety. This variety is called the *Chow variety* of effective r-cycles of degree d on $\mathbb{P}^n$.

EXAMPLE 4.2. Let V be the subspace of homogeneous polynomials in $\mathbb{F}[X_1,\dots,X_{n+1}]$ of degree d. Then $C(n-1,d,n)$ is simply the projective space $\mathbb{P}V$ (see Example 3.9). ◊

EXAMPLE 4.3. $C(r,1,n)$ is the Grassmannian $\mathrm{G}(r+1,n+1)$ (or $\mathbb{G}(r,n)$) embedded in $\mathbb{P}^{\binom{n+1}{r+1}-1} = \mathbb{P}^{\binom{n+1}{k+1}-1}$ via $\phi^\vee$ (see Example 3.10). ◊

The dimension of Chow varieties. When $d = 1$, the Chow variety $C(r,d,n)$ is just the Grassmannian $\mathrm{G}(r+1,n+1)$ (see Example 4.3) and its dimension is well known to be $(r+1)(n-r)$ (Harris 1992). When $d > 1$, the dimension of $C(r,d,n)$ was determined by Azcue (1992) and independently by Lehmann (2017). We state their result as follows.

THEOREM 4.4 (Azcue 1992; Lehmann 2017). *For $d > 1$ and $0 \le r < n$, the dimension of $C(r,d,n)$ is*

$$\max\left\{d(r+1)(n-r), \binom{d+r+1}{r+1} - 1 + (r+2)(n-r-1)\right\}.$$

This theorem was previously proved by Eisenbud & Harris (1992) for the special case $r = 1$.

Remark 4.5. *To prove Theorem 1.10, we only need a lower bound for the dimension of the Chow variety, which is much easier to prove than Theorem 4.4. Indeed, it is not difficult to see that* $d(r+1)(n-r)$ *is the dimension of the space of unions of* d r*-subspaces of* $\mathbb{P}^n$*, and* $\binom{d+r+1}{r+1}-1+(r+2)(n-r-1)$ *is the dimension of the space of degree-*d *hypersurfaces in* $(r+1)$*-subspaces of* $\mathbb{P}^n$*.*

Lower bound via dimension counting. We now restate Theorem 1.10 and prove it using a dimension counting argument.

Theorem 1.10. *Let* $n, d \in \mathbb{N}^+$ *and* $k \in \{0, 1, \ldots, n-1\}$*. Let* $\mathcal{F}$ *be the family of equidimensional projective subvarieties of* $\mathbb{P}^n$ *of dimension* $n-k-1$ *and degree at most* d*. Suppose* $\mathcal{H}$ *is an* $\mathcal{F}$*-evasive* k*-subspace family on* $\mathbb{P}^n$*. Then*

$$|\mathcal{H}| \geq \begin{cases} (n-k)(k+1)+1 & \text{if } d=1, \\ \max\left\{d(n-k)(k+1)+1, \binom{d+n-k}{d}+(n-k+1)k\right\} & \text{if } d>1. \end{cases}$$

In particular, $|\mathcal{H}|$ *is superpolynomial in* n *when* $n-k=n^{\Omega(1)}$ *and* $d=\omega(1)$*.*

Proof. Consider an arbitrary k-subspace $W \in \mathcal{H}$. We may think of each point in $\mathbb{P}(R/I)_d$ as a homogeneous polynomial of degree d in Plücker coordinates modulo scalars and the ideal I of Plücker relations. We know Plücker coordinates always satisfy the Plücker relations. So it makes sense to talk about if a point in $\mathbb{P}(R/I)_d$ vanishes at $\phi(W)$ or not, as it does not depend on the choice of the homogeneous polynomial representing this point. Note that the constraint of $p \in \mathbb{P}(R/I)_d$ vanishing at $\phi(W)$ is a linear equation in the homogeneous coordinates of p. So the set of points in $\mathbb{P}(R/I)_d$ vanishing at $\phi(W)$ is a hyperplane of $\mathbb{P}(R/I)_d$, which we denote by H_W.

Let $r = n-k-1$. Assume $|\mathcal{H}| \leq \dim(C(r,d,n))$. Then we have

$$C(r,d,n) \cap \bigcap_{W \in \mathcal{H}} H_W \neq \emptyset$$

since taking the intersection with a hyperplane reduces the dimension of a projective subvariety by at most one (Lemma 2.7 or

Lemma 2.9). So there exists an effective r-cycle $D = \sum_{i=1}^{s} c_i \mathcal{V}_i \in C(r, d, n)$, where $c_1, \ldots, c_s > 0$, such that $\psi(D) = [\mathcal{R}_D]$ vanishes at $\phi(W)$ for all $W \in \mathcal{H}$.

Let $\mathcal{V} = \operatorname{supp}(D) = \bigcup_{i=1}^{s} \mathcal{V}_i$. Note $\mathcal{V} \in \mathcal{F}$ since $\deg(\mathcal{V}) \leq \deg(D) = d$. For all $W \in \mathcal{H}$, we know $\mathcal{R}_D = \prod_{i=1}^{s} \mathcal{R}_{\mathcal{V}_i}^{c_i}$ vanishes at $\phi(W)$, or equivalently, $\mathcal{R}_{\mathcal{V}} = \prod_{i=1}^{s} \mathcal{R}_{\mathcal{V}_i}$ vanishes at $\phi(W)$. This implies $\mathcal{V} \cap W \neq \emptyset$ for all $W \in \mathcal{H}$. As $\mathcal{V} \in \mathcal{F}$, this contradicts our assumption about $\mathcal{H}$. We conclude

$$|\mathcal{H}| \geq \dim(C(r, d, n)) + 1.$$

The dimension of $C(r, d, n)$ is $(r + 1)(n - r)$ when $d = 1$ and is given by Theorem 4.4 when $d > 1$. Plugging in $r = n - k - 1$ proves the theorem. □

A non-explicit construction. Next, we show that the lower bound in Theorem 1.10 is tight by matching it with a non-explicit construction.

First, we need a bound for the degree of $C(r, d, n)$. Define $M(r, d, n)$ by

$$M(r, d, n) := \begin{cases} ((r + 1)(n - r))! \prod_{i=1}^{r+1} \frac{(i-1)!}{(n-r+i-1)!} & \text{if } d = 1, \\ 3^{\lambda} & \text{if } d > 1. \end{cases}$$

where $\lambda := \min \left\{ \binom{n+d}{d}^{r+1}, \binom{n+d}{d}^{n-r}, \binom{\binom{n+1}{r+1}+d-1}{d} \right\} - 1$.

Lemma 4.6. *The degree of $C(r, d, n)$ in $\mathbb{P}(R/I)_d$ is at most $M(r, d, n)$.*

Proof. When $d = 1$, $C(r, d, n)$ is the Grassmannian $\mathbb{G}(r, n)$ and its degree under the Plücker embedding is known to be exactly $M(r, d, n)$ (Kleiman 1976).

Now assume $d > 1$. In this case, we use the following argument in Catanese (1992). Green & Morrison (1986) proved that the Chow variety $C(r, d, n)$ is defined by equations of degree at most three. It follows from Lemma 2.7 (or Bézout's inequality (Heintz 1983)) that the degree of $C(r, d, n)$ in $\mathbb{P}(R/I)_d$ is bounded by $3^{\dim(\mathbb{P}(R/I)_d)}$. So it remains to prove that $\dim(\mathbb{P}(R/I)_d) \leq \lambda$.

Recall that $R = \mathbb{F}\left[X_S : S \in \binom{[n+1]}{k+1}\right]$ where $k = n - r - 1$. So $\dim(R_d) = \binom{\binom{n+1}{k+1}+d-1}{d} = \binom{\binom{n+1}{r+1}+d-1}{d}$. Therefore,

$$\dim(\mathbb{P}(R/I)_d) \leq \dim(\mathbb{P}R_d) = \binom{\binom{n+1}{r+1}+d-1}{d} - 1.$$

On the other hand, the linear map $\mathcal{R}_\mathcal{V} \mapsto \widetilde{R}_\mathcal{V}$ embeds $\mathbb{P}(R/I)_d$ in $\mathbb{P}V$, where $V \subseteq \mathbb{F}[Y_{i,j} : i \in [k+1], j \in [n+1]]$ is the linear space of multihomogeneous polynomials of degree $(d, \ldots, d)$ in the $k+1$ groups of variables $\{Y_{i_1}, \ldots, Y_{i,n+1}\}$, $i = 1, \ldots, k+1$. Therefore,

$$\dim(\mathbb{P}(R/I)_d) \leq \dim(\mathbb{P}V) = \binom{n+d}{d}^{k+1} - 1 = \binom{n+d}{d}^{n-r} - 1.$$

Similarly, using the linear map $\mathcal{R}^\vee_\mathcal{V} \mapsto \widetilde{R}^\vee_\mathcal{V}$, we get $\dim(\mathbb{P}(R/I)_d) \leq \binom{n+d}{d}^{r+1} - 1$. □

The following theorem gives a non-explicit construction of $\mathcal{H}$ whose cardinality matches the lower bound $\dim(C(r,d,n)) + 1$ in Theorem 1.10.

THEOREM 4.7. *Let $n, d \in \mathbb{N}^+$, $k \in \{0, 1, \ldots, n-1\}$, $r = n - k - 1$, $t = \dim(C(r,d,n))$, and $\delta > 0$. Let $\mathcal{F}$ be the family of equidimensional projective subvarieties of $\mathbb{P}^n$ of dimension $n-k-1$ and degree at most d. Let S be a finite subset of $\mathbb{F}$ such that*

$$|S| \geq M(r,d,n) \cdot t(k+1)d/\delta.$$

Let $\mathcal{H} = \{W_1, \ldots, W_{t+1}\} \subseteq \mathbb{G}(k,n)$ where the entries of the generating matrices of $W_1, \ldots, W_{t+1}$ are chosen independently at random from S. Then with probability at least $1 - \delta$, $\mathcal{H}$ is an $\mathcal{F}$-evasive k-subspace family on $\mathbb{P}^n$.

PROOF. Whenever $\mathcal{H}$ fails to be $\mathcal{F}$-evasive, there exists an effective r-cycle D of degree at most d such that $\mathrm{supp}(D)$ intersects W for all $W \in \mathcal{H}$. By adding extra r-subspaces to D if necessary, we may assume the degree of D is exactly d, i.e., $D \in C(r,d,n)$.

As argued in the proof of Theorem 1.10, for a k-subspace $W \in \mathbb{G}(k,n)$, the support of $D \in C(r,d,n)$ intersects W iff D lies in a

hyperplane H_W of $\mathbb{P}(R/I)_d$ corresponding to W. So we just need to prove that the condition

$$C(r,d,n) \cap \bigcap_{W\in\mathcal{H}} H_W = \emptyset$$

holds with probability at least $1-\delta$. Suppose $W_1,\ldots,W_{i-1}$ are already chosen. Let $C = C(r,d,n) \cap \bigcap_{j=1}^{i-1} H_{W_j}$. By induction, it suffices to show that $\dim(C \cap H_{W_i}) \le \dim(C) - 1$ holds with probability at least $1-\delta/t$. (Again, the dimension of an empty set is assumed to be $-\infty$.)

Consider an irreducible component C_0 of C. Fix $D = \sum_{j=1}^{s} c_j \mathcal{V}_j \in C_0$. The Chow form $\widetilde{R}_D = \prod_{j=1}^{s} \widetilde{R}_{\mathcal{V}_j}^{c_j}$ is a nonzero polynomial of degree $(k+1)d$. Let M be the randomly chosen generating matrix of W_i. By the Schwartz–Zippel Lemma (Schwartz 1980; Zippel 1979), $\widetilde{R}_D(M) \neq 0$ holds with probability at least $1-(k+1)d/|S|$. When this occurs, we have $D \notin H_{W_i}$ and hence $\dim(C_0 \cap H_{W_i}) \le \dim(C_0) - 1$ by Lemma 2.7. The number of irreducible components of $C = C(r,d,n) \cap \bigcap_{j=1}^{i-1} H_{W_j}$ is bounded by $\deg(C) \le \deg(C(r,d,n)) \le M(r,d,n)$, where the first inequality uses Lemma 2.7 and the second inequality holds by Lemma 4.6. By the union bound, the probability that $\dim(C \cap H_{W_i}) \le \dim(C) - 1$ does not occur is bounded by

$$M(r,d,n) \cdot (k+1)d/|S| \le \delta/t$$

as desired. □

Remark 4.8. *While the cardinality $|\mathcal{H}|$ in Theorem 4.7 is optimal, an unsatisfying issue here is that the elements in S are huge when $d > 1$. In particular, when $\min\{k, n-k\}$ is linear in n, these elements have exponential bit-length even if $d > 1$ is bounded. This is due to the poor bound $M(r,d,n)$ for the number of irreducible components that we use. We suspect that this bound can be greatly improved. Kollár (1996, Exercise 3.28) outlined a method of proving a more effective bound for the number of irreducible components of $C(r,d,n)$. Guerra (1999) extended this method for Chow varieties associated with general projective varieties. Unfortunately, it is not clear to us if this method can be*

extended to bound the number of irreducible components of the intersection $C(r,d,n) \cap \bigcap_{j=1}^{i-1} H_{W_j}$. *So we leave it as an open problem to obtain a more effective bound for the entries of the generating matrices in Theorem 4.7.*

5. Application to PIT for Depth-4 Circuits

In this section, we use explicit variety evasive subspace families to obtain a black-box PIT algorithm for non-SG $\Sigma\Pi\Sigma\Pi(k,r)$ circuits, thereby proving Theorem 1.15. The proof only uses the simple construction of variety evasive subspace families (Lemma 3.11).

We first define $\Sigma\Pi\Sigma\Pi(k,r)$ circuits and non-SG $\Sigma\Pi\Sigma\Pi(k,r)$ circuits.

Definition 5.1 ($\Sigma\Pi\Sigma\Pi(k,r)$ circuit). *An algebraic circuit* C *over* $\mathbb{F}$ *is a* $\Sigma\Pi\Sigma\Pi(k,r)$ circuit *if it has the form*

$$C(X_1,\ldots,X_n) = \sum_{i=1}^{k'} F_i = \sum_{i=1}^{k'} \prod_{j=1}^{d_i} Q_{i,j} \tag{5.2}$$

where $k' \le k$, $d_1,\ldots,d_{k'} \in \mathbb{N}^+$, $F_i = \prod_{j=1}^{d_i} Q_{i,j}$ *for* $i \in [k']$, *and each* $Q_{i,j}$ *is a polynomial in* $X_1,\ldots,X_n$ *of degree at most* r *over* $\mathbb{F}$. *The* degree *of the circuit* C *is defined to be* $\max\{\deg(F_i) : i \in [k']\}$. *In addition:*

- C *is* minimal *if* $\sum_{i\in I} F_i \neq 0$ *for all nonempty proper subset* $I \subseteq [k']$.
- C *is* homogeneous *if all the polynomials* F_i *are homogeneous of the same degree.*
- *Let* $\gcd(C) := \gcd(F_1,\ldots,F_{k'})$. *We say* C *is* simple *if* $\gcd(C) = 1$. *In general, we have* $C = \gcd(C)\cdot \mathrm{sim}(C)$ *where* $\mathrm{sim}(C)$ *is a simple* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit, called the* simple part *of* C. *Note the simple part of a minimal* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit is still minimal.*

The polynomial computed by C *is again denoted by* C *by an abuse of notation.*

Definition 5.3 (Non-SG circuit). *We say a minimal, simple, and homogeneous* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit* $C(X_1,\dots,X_n)=\sum_{i=1}^{k'}F_i$ *as in (5.2) is* non-SG *if there exists* $i\in[k']$ *such that*

$$\bigcap_{j\in[k']\setminus i}\mathcal{V}(F_j)\not\subseteq\mathcal{V}(F_i)$$

where $\mathcal{V}(F)$ *denotes the subvariety of* $\mathbb{P}^n$ *defined by* F*. More generally, a minimal and simple* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit* $C(X_1,\dots,X_n)=\sum_{i=1}^{k'}F_i$ *of degree* d *is non-SG if its homogenization*

$$\begin{aligned}\widetilde{C}(X_1,\dots,X_{n+1})&=\sum_{i=1}^{k'}F_i(X_1/X_{n+1},\dots,X_n/X_{n+1})\cdot X_{n+1}^d\\&=\sum_{i=1}^{k'}\prod_{j=1}^{d_i'}\widetilde{Q}_{i,j}\end{aligned}$$

is non-SG, where $d_i'=d_i+(d-\deg(F_i))$ *and* $\widetilde{Q}_{i,j}$ *equals the homogenization of* $Q_{i,j}$ *if* $j\le d_i$ *and equals* X_{n+1} *if* $j>d_i$*. A minimal* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit* C *is non-SG if* $\operatorname{sim}(C)$ *is non-SG. Finally, a* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit is non-SG if it has an equivalent minimal non-SG* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuit.*

We restate our result (Theorem 1.15) and then give a proof.

Theorem 1.15. *There exists a deterministic black-box PIT algorithm with time complexity polynomial in* $d\cdot\binom{k(n+1+r^k)}{kr^k}\cdot\binom{k-1+d}{k-1}\le\operatorname{poly}(d^k,n^{r^k},r^{k^2r^k})$ *(and* $\log p$*, if* $\operatorname{char}(\mathbb{F})=p>0$*) for non-SG* $\Sigma\Pi\Sigma\Pi(k,r)$ *circuits of degree at most* d *in* $X_1,\dots,X_n$ *over an algebraically closed field* $\mathbb{F}$*.*

Proof. If $n\le k-1$, we may simply use Lemma 2.3 to construct a $\frac{1}{2}$-hitting set of size polynomial in $\binom{n+d}{n}\le\binom{k-1+d}{k-1}$ for n-variate polynomials of degree at most d, and then run the corresponding black-box PIT algorithm. So assume $n>k-1$.

Consider a nonzero non-SG $\Sigma\Pi\Sigma\Pi(k,r)$ circuit C of degree at most d. We want to design a black-box PIT algorithm for C. By replacing C with an equivalent minimal non-SG circuit, we may

assume C is minimal. Let $D = \gcd(C)$ and $E = \mathrm{sim}(C)$. Let $\widetilde{C}$, $\widetilde{D}$, and $\widetilde{E}$ be the homogenization of C, D, and E respectively. Then $\widetilde{D} = \gcd(\widetilde{C})$, $\widetilde{E} = \mathrm{sim}(\widetilde{C})$, and $\widetilde{C} = \widetilde{D} \cdot \widetilde{E}$.

Let $\mathcal{H}$ be an affine $(k-1)$-subspace family on $\mathbb{A}^n$ of size poly $(\binom{k(n+1+r^k)}{kr^k}, d)$ such that $\mathcal{H}' := \{W_{\mathrm{cl}} : W \in \mathcal{H}\}$ is an $(n, r^k, \frac{1}{4d})$-evasive $(k-1)$-subspace family on $\mathbb{P}^n$. Such a family $\mathcal{H}$ can be computed using Lemma 3.15 and Lemma 3.11. We claim

1. $\widetilde{D}|_W \neq 0$ for all but at most $\frac{1}{4}$-fraction of $W \in \mathcal{H}'$, and
2. $\widetilde{E}|_W \neq 0$ for all but at most $\frac{1}{4}$-fraction of $W \in \mathcal{H}'$.

Assume these two claims hold. Then for at least half of $W \in \mathcal{H}$, we have $\widetilde{C}|_{W_{\mathrm{cl}}} \neq 0$ and hence $C|_W = \widetilde{C}|_{W_{\mathrm{cl}} \cap \mathbb{A}^n} \neq 0$, where we use the facts that $\widetilde{C}(X_1, \ldots, X_n, 1)$ equals $C(X_1, \ldots, X_n)$ and $W_{\mathrm{cl}} \cap \mathbb{A}^n$ is dense in W_{cl}. The restriction of C to each $W \cong \mathbb{A}^{k-1}$ is a $(k-1)$-variate polynomial of degree at most d. So to test if $C|_W$ is zero, we just need to use Lemma 2.3 to construct a hitting set in W of size poly$(\binom{k-1+d}{k-1})$ for $(k-1)$-variate polynomials of degree at most d. Take the union of these hitting sets to obtain a hitting set of size poly$(\binom{k(n+1+r^k)}{kr^k}, d, \binom{k-1+d}{k-1})$ and we are done.

So it remains to prove the two claims. Note $\widetilde{D}$ is the product of at most d factors whose degrees are bounded by r. The first claim then follows from the $(n, r^k, \frac{1}{4d})$-evasiveness of $\mathcal{H}'$ and the union bound.

Now we prove the second claim. By definition, $\widetilde{E}$ is a non-SG $\Sigma\Pi\Sigma\Pi(k, r)$ circuit. Suppose it has the form

$$\widetilde{E} = \sum_{i=1}^{k'} F_i = \sum_{i=1}^{k'} \prod_{j=1}^{d_i} Q_{i,j} \tag{5.4}$$

where each $Q_{i,j}$ is a homogeneous polynomial of degree at most r. As $\widetilde{E}$ is non-SG, there exists $i_0 \in [k']$ such that

$$\bigcap_{i \in [k'] \setminus i_0} \mathcal{V}(F_i) \not\subseteq \mathcal{V}(F_{i_0})$$

Without loss of generality, we may assume $i_0 = k'$. Note $\mathcal{V}(F_i) = \bigcup_{j=1}^{d_i} \mathcal{V}(Q_{i,j})$ for $i \in [k']$. So there exists $(j_1, \ldots, j_{k'-1}) \in [d_1] \times$

$\cdots \times [d_{k'-1}]$ such that

$$\bigcap_{i=1}^{k'-1} \mathcal{V}(Q_{i,j_i}) \not\subseteq \mathcal{V}(F_{k'}).$$

Let $\mathcal{V}_0$ be an irreducible component of $\bigcap_{i=1}^{k'-1} \mathcal{V}(Q_{i,j_i})$ such that $\mathcal{V}_0 \not\subseteq \mathcal{V}(F_{k'})$. Let $d_0 = \dim(\mathcal{V}_0) \geq 0$. By Lemma 2.7, we have $d_0 \geq n - k' + 1$ and the variety $\mathcal{V}_0 \cap \mathcal{V}(F_{k'}) = \bigcup_{j=1}^{d_{k'}} (\mathcal{V}_0 \cap \mathcal{V}(Q_{k',j}))$ has dimension at most $d_0 - 1$. For each $j \in [d_{k'}]$, the degree of $\mathcal{V}_0 \cap \mathcal{V}(Q_{k',j})$ is at most r^k by Lemma 2.7 (or by Bézout's inequality (Heintz 1983)). By $(n, r^k, \frac{1}{4d})$-evasiveness of $\mathcal{H}'$ and the union bound, all but at most $\frac{1}{4}$-fraction of $W \in \mathcal{H}'$ evade $\mathcal{V}_0 \cap \mathcal{V}(Q_{k',j})$ for $j = 1, 2, \ldots, d_{k'}$.

Consider any $W \in \mathcal{H}'$ that evades $\mathcal{V}_0 \cap \mathcal{V}(Q_{k',j})$ for $j = 1, 2, \ldots, d_{k'}$. We just need to prove $\widetilde{E}|_W \neq 0$, or equivalently, $W \not\subseteq \mathcal{V}(\widetilde{E})$. Assume to the contrary that $W \subseteq \mathcal{V}(\widetilde{E})$. Then $W \cap \mathcal{V}_0 \subseteq \mathcal{V}(\widetilde{E})$. So

$$\begin{aligned} W \cap \mathcal{V}_0 = W \cap \mathcal{V}_0 \cap \mathcal{V}(\widetilde{E}) &= W \cap \mathcal{V}_0 \cap \mathcal{V}\left(\prod_{j=1}^{d_{k'}} Q_{k',j}\right) \\ &= \bigcup_{j=1}^{d_{k'}} (W \cap \mathcal{V}_0 \cap \mathcal{V}(Q_{k',j})) \end{aligned} \tag{5.5}$$

where the second equality holds since $\widetilde{E} \equiv \prod_{j=1}^{d_{k'}} Q_{k',j}$ modulo the ideal $I_0 := \langle Q_{1,j_1}, \ldots, Q_{k'-1,j_{k'-1}} \rangle$ by (5.4) and $\mathcal{V}_0 \subseteq \bigcap_{i=1}^{k'-1} \mathcal{V}(Q_{i,j_i}) = \mathcal{V}(I_0)$. We know the dimension of $\bigcup_{j=1}^{d_{k'}} (\mathcal{V}_0 \cap \mathcal{V}(Q_{k',j}))$ is at most $d_0 - 1$. So by the choice of W, the dimension of $\bigcup_{j=1}^{d_{k'}} (W \cap \mathcal{V}_0 \cap \mathcal{V}(Q_{k',j}))$ is at most $(k-1) + (d_0 - 1) - n$. However, by Lemma 2.9, the dimension of $W \cap \mathcal{V}_0$ is at least $(k-1) + d_0 - n \geq 0$, where we use the fact $d_0 \geq n - k' + 1 \geq n - k + 1$. This contradicts (5.5). So $\widetilde{E}|_W \neq 0$. □

6. Open Problems and Future Directions

We have seen that constructing explicit variety evasive subspace families is a natural problem that generalizes important problems

in algebraic pseudorandomness and algebraic complexity theory, including deterministic black-box polynomial identity testing (evading varieties of codimension one) and constructing explicit lossless rank condensers (evading varieties of degree one). It is closely connected with advanced topics in algebraic geometry such as Chow forms and Chow varieties, and has applications to derandomizing PIT and non-explicit results in algebraic geometry like Noether's normalization lemma.

There are many interesting open problems and potential future directions. We list some of them here.

1. Theorem 1.7 focuses on subvarieties of bounded degree in a projective or affine space. Are there other interesting families of varieties for which we could construct explicit variety evasive subspace families? Families that are defined computation-theoretically may be particularly interesting, as many results of this kind are already known for polynomial identity testing.

2. Can explicit variety evasive subspace families be used to derandomize other non-explicit results in algebraic geometry?

3. Can our explicit construction in Theorem 1.7 be improved? In the case $k = 0$ and the case $d = 1$, there are optimal or essentially optimal constructions, and our construction indeed degenerates into these constructions. In general, however, there is a significant gap between the upper bound in Theorem 1.7 and the lower bound in Theorem 1.10.

4. Extending the notion of *strong* lossless rank condensers (Forbes & Guruswami 2015), one could strengthen the definition of $(\mathcal{F}, \varepsilon)$-evasive subspace families in Definition 1.3 by bounding the total deviation of the dimension instead of the number of bad subspaces. At the same time, one could consider the setting where there is a gap between $\dim(\mathcal{V}_1)$ and $\operatorname{codim}(\mathcal{V}_2)$, as in typical applications of *subspace designs* (Guo & Ron-Zewi 2021; Guruswami *et al.* 2021; Guruswami & Xing 2013). Alternatively, one could relax the definition by allowing $\dim(\mathcal{V}_1 \cap \mathcal{V}_2)$ to be slightly greater than $\dim(\mathcal{V}_1) + \dim(\mathcal{V}_2) - n$, which

is related to the notion of *lossy* rank condensers in (Forbes & Guruswami 2015). It is natural to study explicit constructions of these variants and their applications, which can be seen as extensions of the theory of "linear-algebraic pseudorandomness" (Forbes & Guruswami 2015) to a nonlinear setting.

5. Could our lower bound (Theorem 1.10) be extended to the affine case or to a "lossy" relaxation of the problem?

6. Is there a more effective bound for the entries of the generating matrices that are used in the non-explicit construction (Theorem 4.7)?

7. When $n-k = O(1)$, our lower bound (Theorem 1.10) is only polynomial in n and d. So one question is if there are explicit constructions of polynomial size when $n - k = O(1)$. As a concrete special case, consider the problem of constructing an explicit affine $(n-2)$-subspace family $\mathcal{H}$ on $\mathbb{A}^n$ such that $\mathcal{H}$ is evasive for degree-d curves that are images of morphisms $\mathbb{A}^1 \to \mathbb{A}^n$. Note that for $\varphi : \mathbb{A}^1 \to \mathbb{A}^n$ corresponding to a ring homomorphism $\varphi^\sharp : \mathbb{F}[X_1, \ldots, X_n] \to \mathbb{F}[Y]$, an affine $(n-2)$-subspace defined by affine linear polynomials ℓ_1 and ℓ_2 evades the curve $\mathrm{Im}(\varphi)$ iff $\varphi^\sharp(\ell_1)$ and $\varphi^\sharp(\ell_2)$ have no common root. Using *resultants*, we could reduce this problem to black-box PIT for symbolic determinants. Unconditionally, Theorem 1.7 also yields an explicit construction of polynomial size when $d = O(\log n)$. We are not aware of any unconditional derandomization whose time complexity is subexponential in $\min\{n, d\}$, however.

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